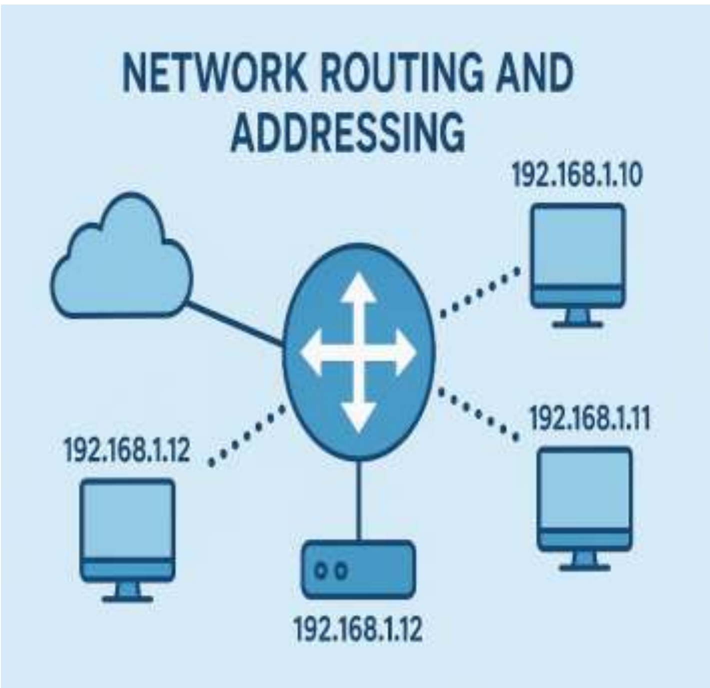


ROUTING AND ADDRESSING

Hands-on-Labs

ABSTRACT

This module focuses on networking in Google Cloud, specifically on routing, addressing, and traffic management. It covers advanced DNS configuration using geolocation-based traffic steering to optimize user experience and direct traffic efficiently. Additionally, it explores enabling secure private connectivity to Google services through Private Google Access and configuring Cloud NAT to allow outbound internet access for resources without public IP addresses. These labs provide practical experience in designing scalable, secure, and geographically-aware network architectures in Google Cloud.

DEBORAH BINYANYA

Contents

CLOUD DNS - TRAFFIC STEERING USING GEOLOCATION POLICY

Overview

Cloud DNS routing policies enable users to configure DNS based traffic steering. A user can either create a **Weighted Round Robin (WRR)** routing policy or a **Geolocation (GEO) routing** policy. You can configure routing policies by creating special ResourceRecordSets with special routing policy values.

Use WRR to specify different weights per ResourceRecordSet for the resolution of domain names. Cloud DNS routing policies help ensure that traffic is distributed across multiple IP addresses by resolving DNS requests according to the configured weights.

In this lab, you will configure and test the Geolocation routing policy. Use GEO to specify source geolocations and to provide DNS answers corresponding to those geographies. The geolocation routing policy applies the nearest match for the source location when the traffic source location doesn't match any policy items exactly.

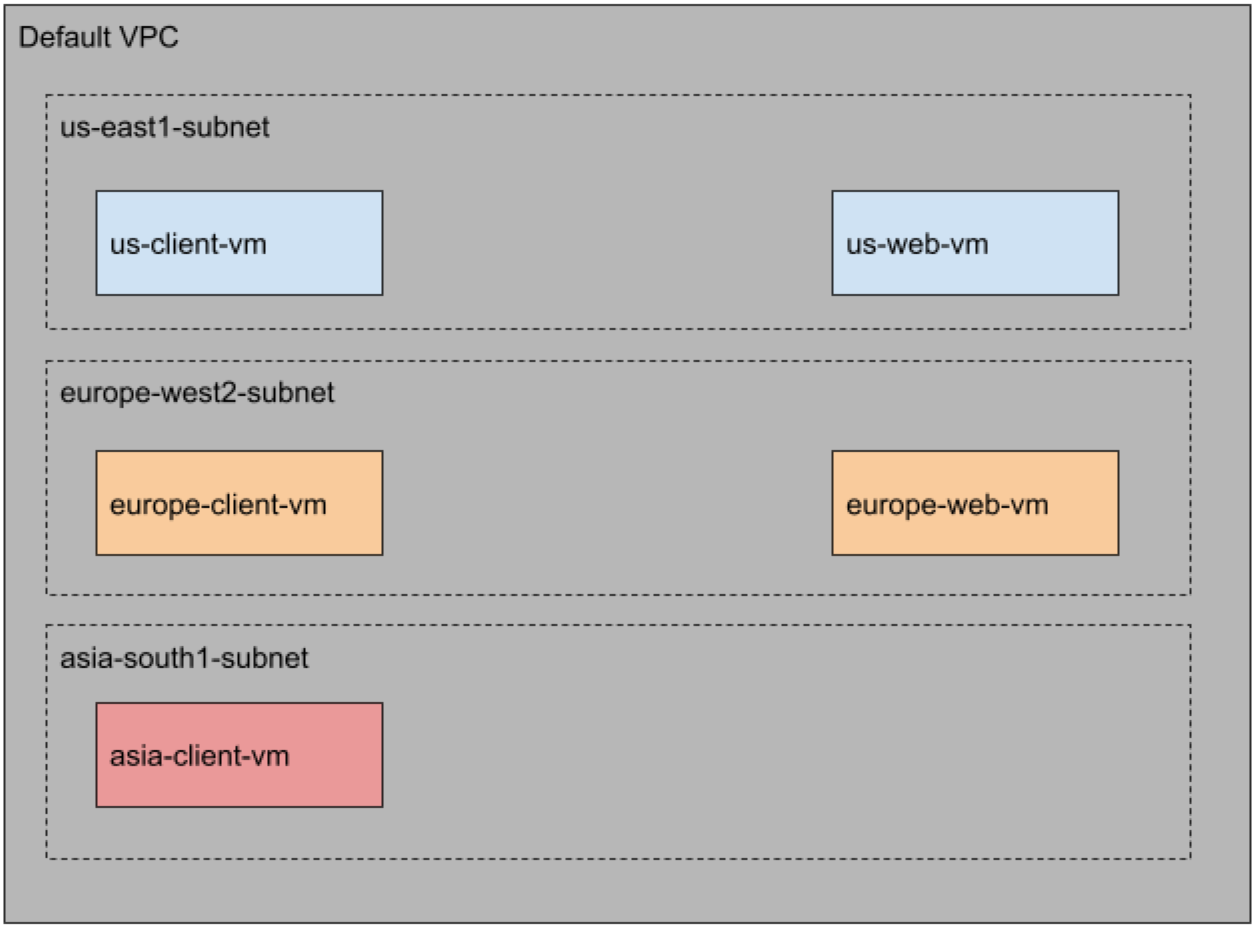
What you learn

You will learn how to:

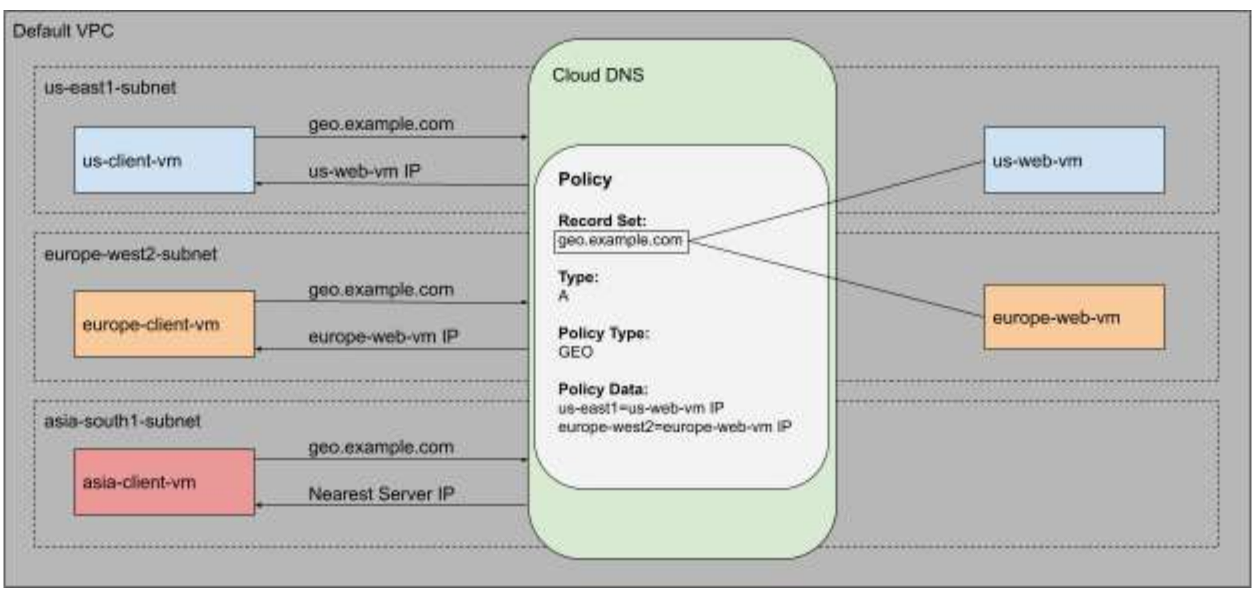
1. Launch client VMs, one in each region
2. Launch server VMs, one in each region except asia-south1
3. Create a private zone, for example.com
4. Create a Geolocation routing policy using gcloud commands
5. Test the configuration

Architecture

Use the default VPC network to create all the virtual machines (VM) and launch client VMs in 3 Google Cloud locations: one in the United States, another in Europe, and another in Asia. To demonstrate the Geolocation routing policy behavior, you will create the server VMs only in two of those location - in the United States and in Europe. The archirtecture will look similar to what is shown in the graphic. (Note that the actual regions and zones within the United States and Europe may differ from those shown in the graphic.)



You will use Cloud DNS routing policies and create ResourceRecordSets for geo.example.com and configure the Geolocation policy to help ensure that a client request is routed to a server in the client's closest region.



Setup and requirements

Before you click the Start Lab button

Note: Read these instructions.

Labs are timed and you cannot pause them. The timer, which starts when you click **Start Lab**, shows how long Google Cloud resources will be made available to you.

This Qwiklabs hands-on lab lets you do the lab activities yourself in a real cloud environment, not in a simulation or demo environment. It does so by giving you new, temporary credentials that you use to sign in and access Google Cloud for the duration of the lab.

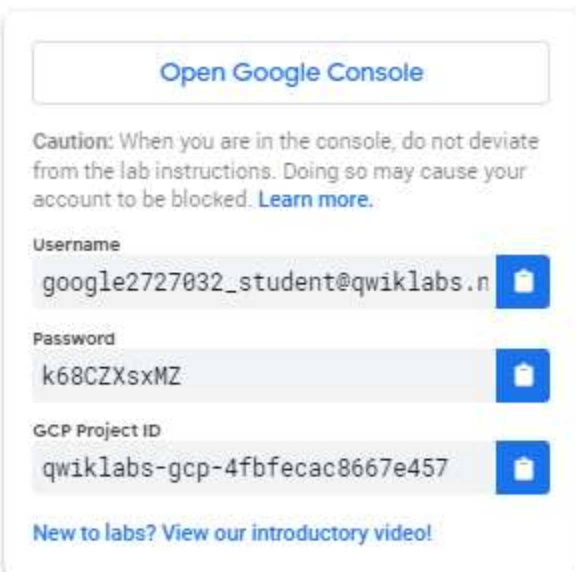
What you need

To complete this lab, you need:

- Access to a standard internet browser (Chrome browser recommended).
- Time to complete the lab.
  Note: If you already have your own personal Google Cloud account or project, do not use it for this lab.**Note:** If you are using a Pixelbook, open an Incognito window to run this lab.

How to start your lab and sign in to the Console

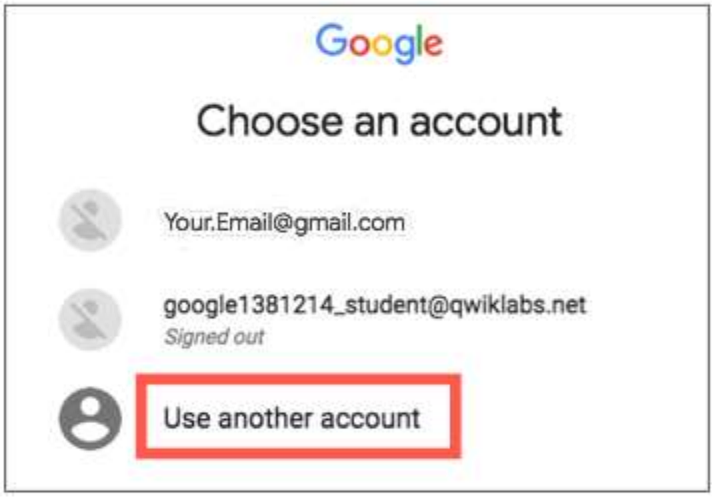
1. Click the **Start Lab** button. If you need to pay for the lab, a pop-up opens for you to select your payment method. On the left is a panel populated with the temporary credentials that you must use for this lab.



2. Copy the username, and then click **Open Google Console**. The lab spins up resources, and then opens another tab that shows the **Choose an account** page.

   Note: Open the tabs in separate windows, side-by-side.

3. On the Choose an account page, click **Use Another Account**. The Sign in page opens.

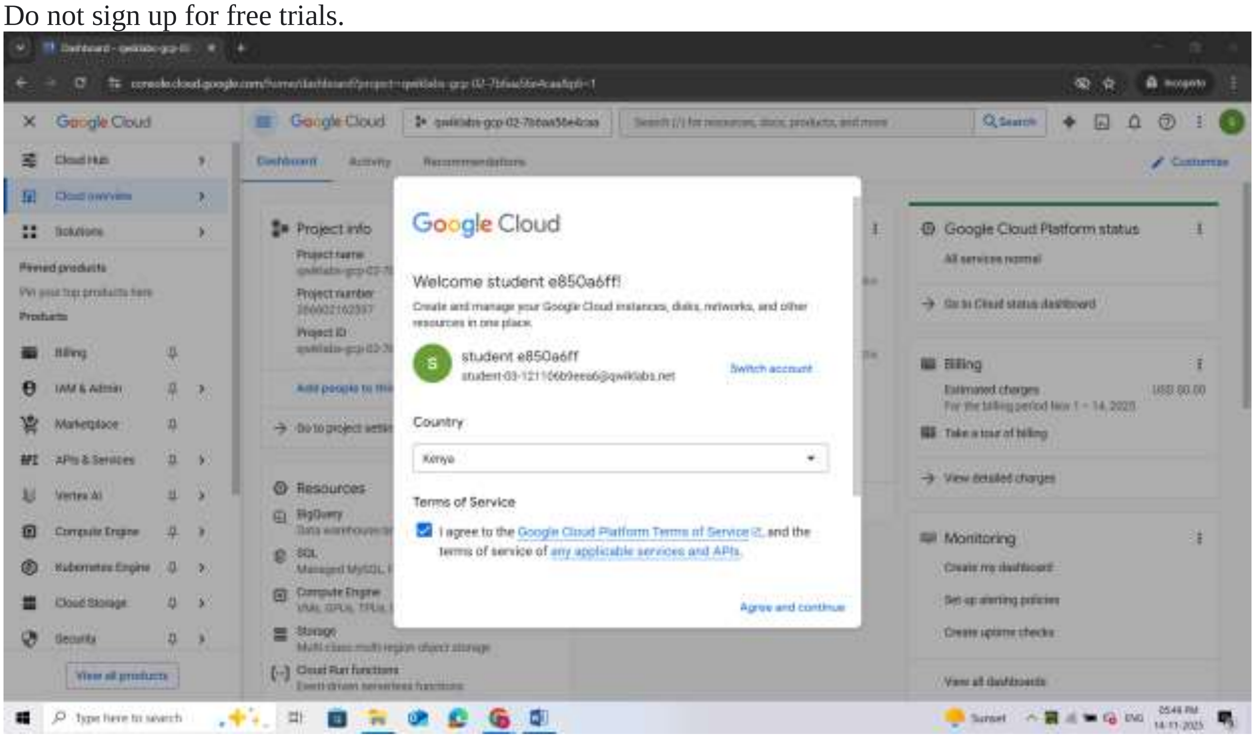


4. Paste the username that you copied from the Connection Details panel. Then copy and paste the password.

Note: You must use the credentials from the Connection Details panel. Do not use your Google Cloud Skills Boost credentials. If you have your own Google Cloud account, do not use it for this lab (avoids incurring charges).

5. Click through the subsequent pages:

- Accept the terms and conditions.
- Do not add recovery options or two-factor authentication (because this is a temporary account).
- Do not sign up for free trials.



After a few moments, the Cloud console opens in this tab.

Note: You can view the menu with a list of Google Cloud Products and Services by clicking the **Navigation menu** at the top-left.

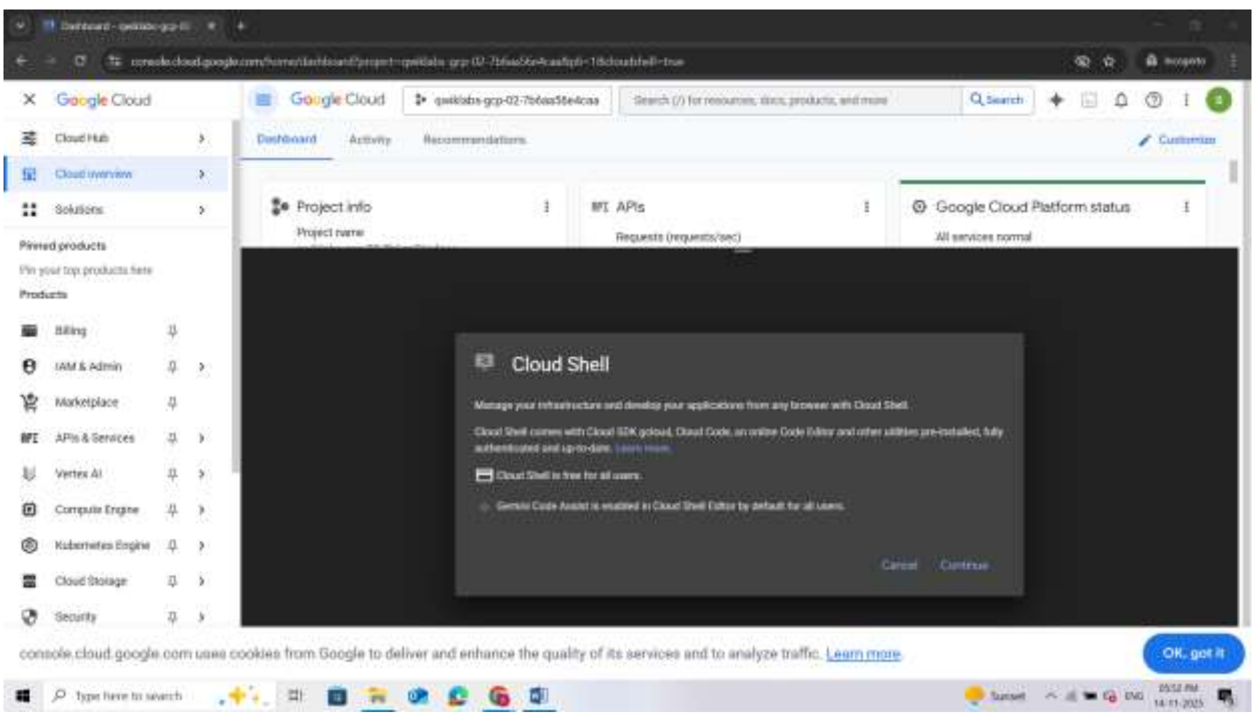
Activate Google Cloud Shell

Google Cloud Shell is a virtual machine that is loaded with development tools. It offers a persistent 5GB home directory and runs on the Google Cloud.

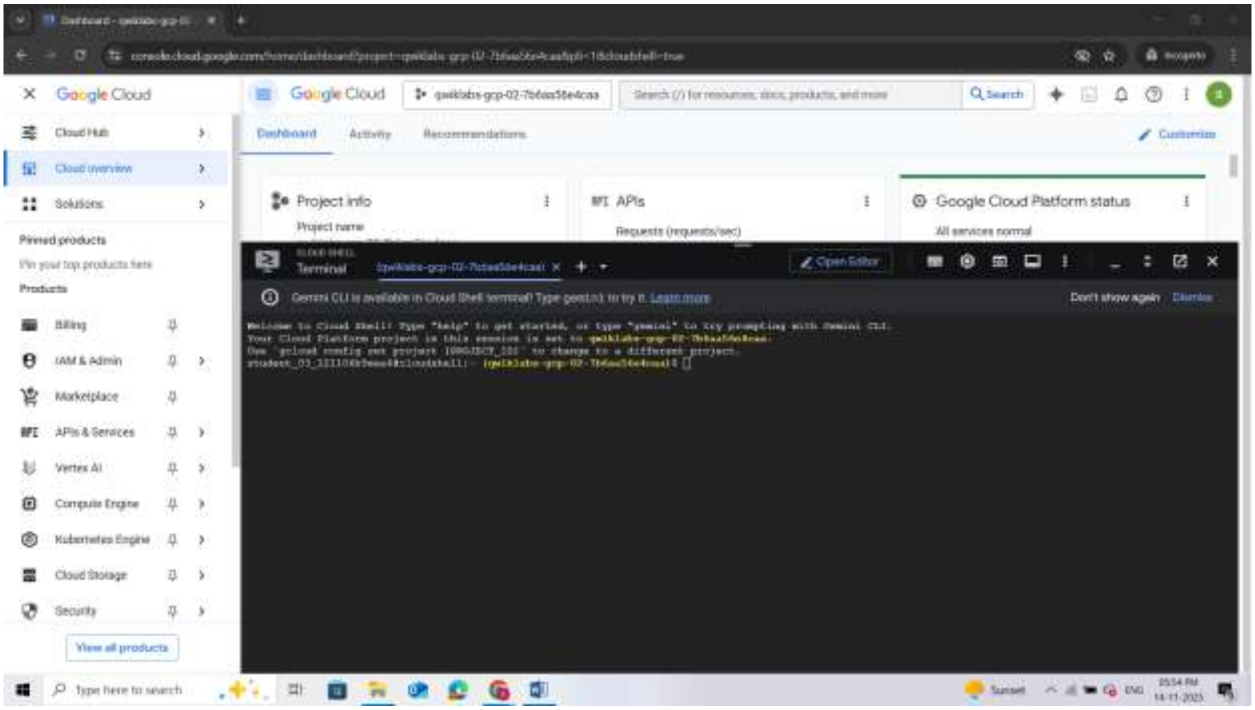
Google Cloud Shell provides command-line access to your Google Cloud resources.

1. In Cloud console, on the top right toolbar, click the Open Cloud Shell button.

2. Click **Continue**.

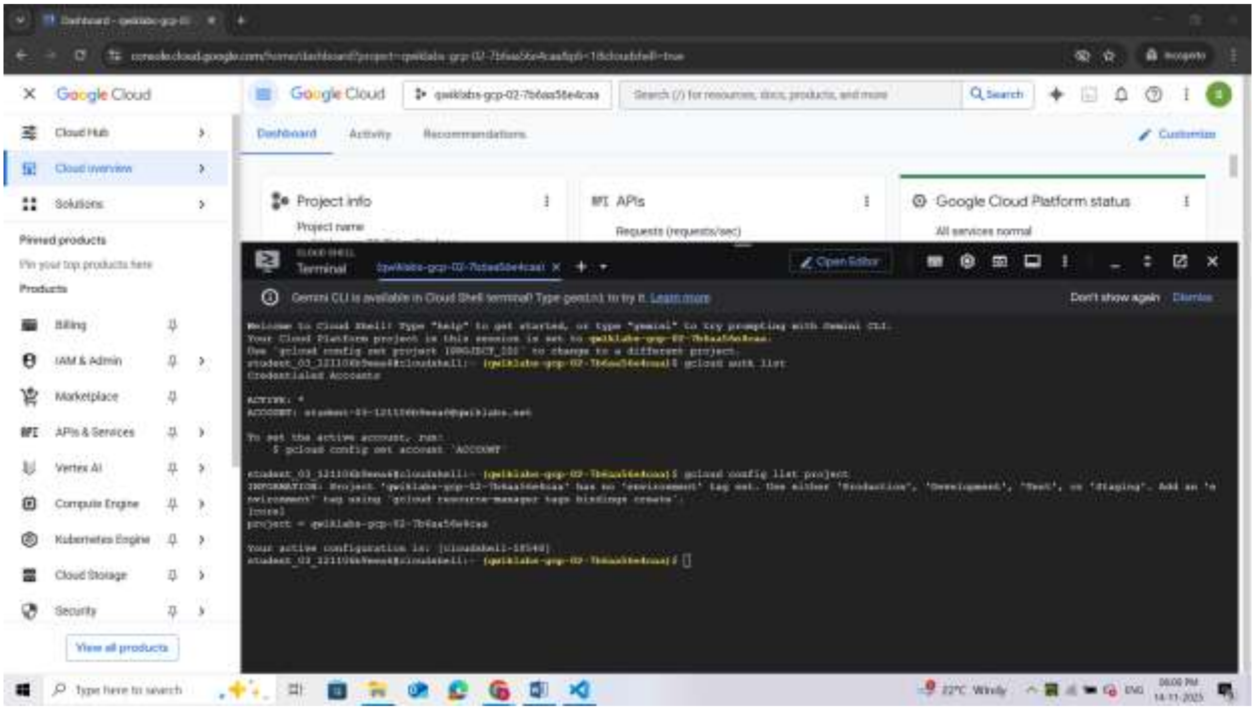


It takes a few moments to provision and connect to the environment. When you are connected, you are already authenticated, and the project is set to your *PROJECT_ID*. For example:



gcloud is the command-line tool for Google Cloud. It comes pre-installed on Cloud Shell and supports tab-completion.

- You can list the active account name with this command:
  gcloud auth list
- You can list the project ID with this command:
  gcloud config list project
  Note: Full documentation of **gcloud** is available in the [gcloud CLI overview guide].



Task 1. Enable APIs

Ensure that the Compute and the Cloud DNS APIs are enabled. In this section, you will enable the APIs manually, using gcloud commands.

Enable Compute Engine API

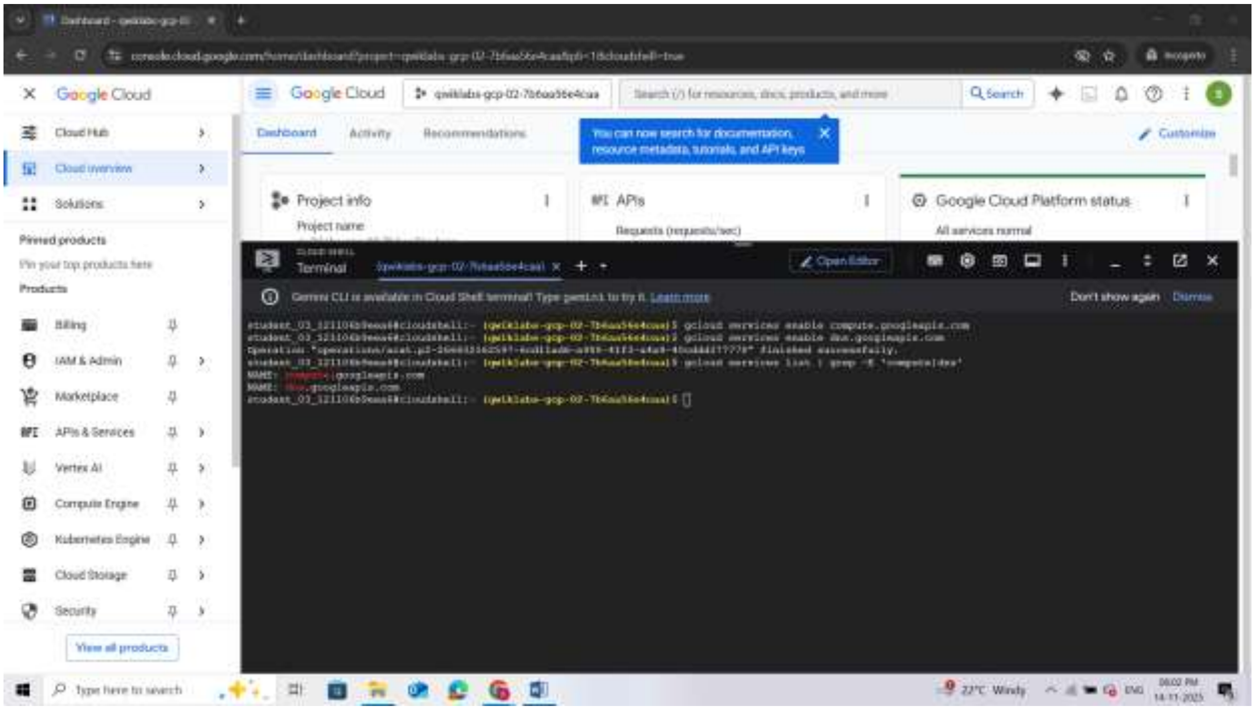
- Run the gcloud services enable command to enable the Compute Engine API:
  gcloud services enable compute.googleapis.com
  This command can take a few minutes to complete.

Enable Cloud DNS API

- Run the gcloud services enable command to enable the Cloud DNS API:
  gcloud services enable dns.googleapis.com
  This command can take a few minutes to complete.

Verify that the APIs are enabled

- Run the gcloud services list command to list all the enabled APIs. We should see compute.googleapis.com and dns.googleapis.com in the listed output.
  gcloud services list | grep -E 'compute|dns'



Task 2. Configure the firewall

Before you create the client VMs and the web servers, you need to create two firewall rules.

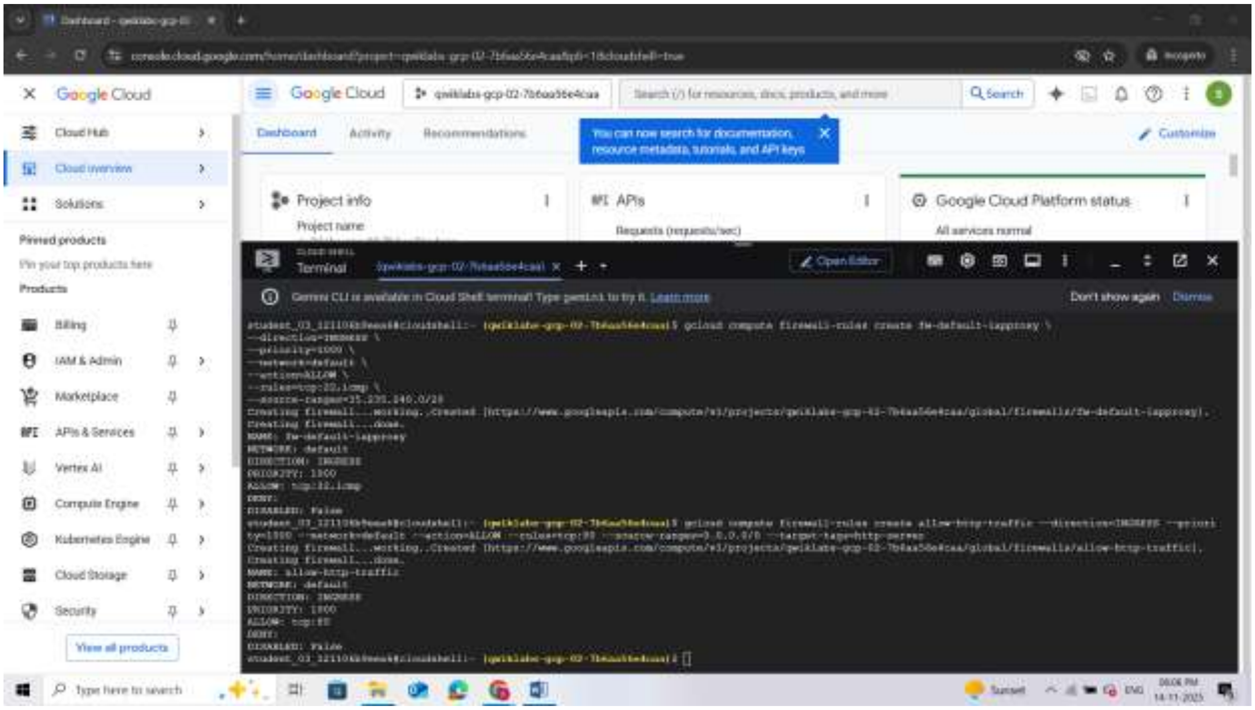
***Note:** The firewall-rules create command can take a few minutes to complete. Please wait for the "Creating firewall...done" message before proceeding to the next step.*

1. To be able to SSH into the client VMs, run the following to create a firewall rule to allow SSH traffic from Identity Aware Proxies (IAP):

```
gcloud compute firewall-rules create fw-default-iapproxy \
--direction=INGRESS \
--priority=1000 \
--network=default \
--action=ALLOW \
--rules=tcp:22,icmp \
--source-ranges=35.235.240.0/20
```

2. To allow HTTP traffic on the web servers, each web server will have a "http-server" tag associated with it. You will use this tag to apply the firewall rule only to your web servers:

```
gcloud compute firewall-rules create allow-http-traffic --direction=INGRESS --priority=1000 --network=default --action=ALLOW --rules=tcp:80 --source-ranges=0.0.0.0/0 --target-tags=http-server
```



Task 3. Launch client VMs

Now that the APIs are enabled, and the firewall rules are in place, the next step is to set up the environment. In this section, you will create 3 client VMs, one in each region.

Launch a client in the United States

- Run the gcloud compute instances create command to create the client VMs:
  gcloud compute instances create us-client-vm --machine-type=e2-micro --zone "Zone 1"
  This command can take a few minutes to complete. Please wait for a "Created" message before moving to the next step. Note that you may see a different zone in gcloud than in the sample output shown below.

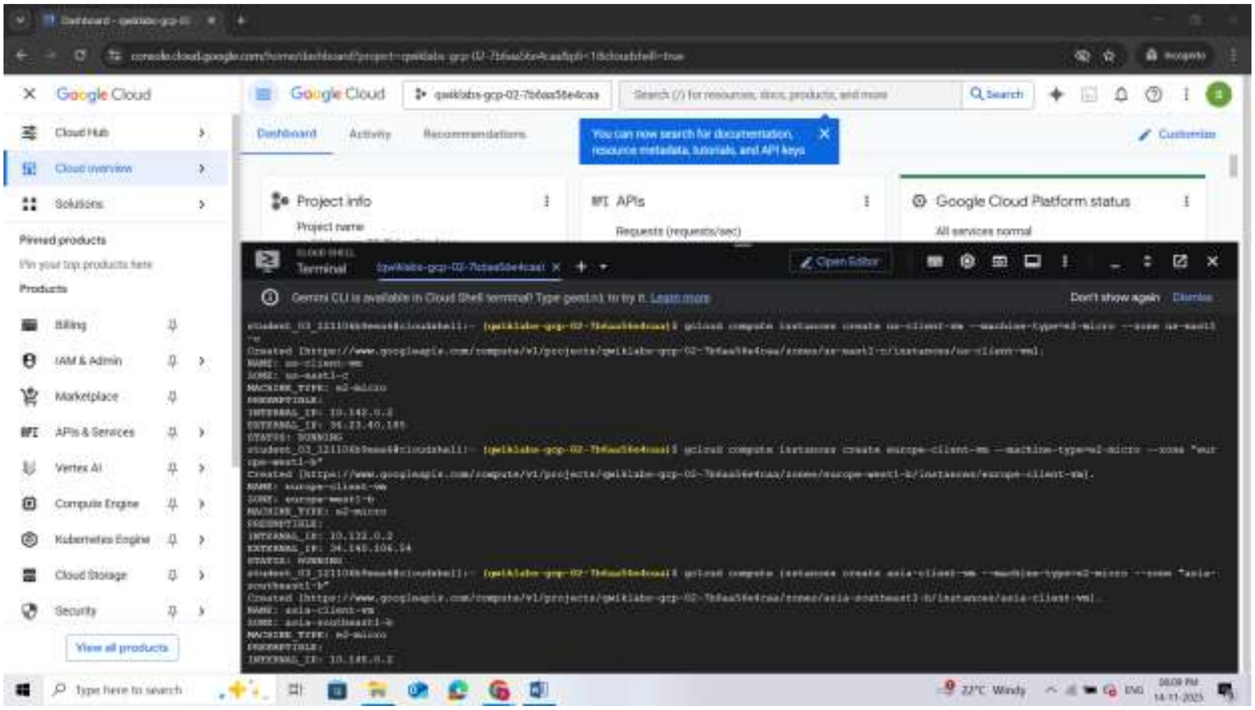
Launch a client in Europe

- Run the following to create the client VMs:
  gcloud compute instances create europe-client-vm --machine-type=e2-micro --zone ""Zone 2""
  Note that you may see a different zone in gcloud than in the sample output shown below.

Launch a client in Asia

1. Run the following to create the client VMs:

gcloud compute instances create asia-client-vm --machine-type=e2-micro --zone ""Zone 3""
Note that you may see a different zone in gcloud than in the sample output shown below.



Task 4. Launch Server VMs

Now that the client VM's are up and running, the next step is to create the server VMs. You will use a startup script to configure and set up the web servers. As mentioned earlier, you will create the server VMs only in 2 regions: us-east1 and europe-west2.

- Run the gcloud compute instances create command to create the server VMs. The compute instance create command can take a few minutes to complete. Please wait for a "Created" message before moving to the next step.

Launch server in the United States

- Run the following command:

```
gcloud compute instances create us-web-vm \
--machine-type=e2-micro \
--zone="Zone 1" \
--network=default \
--subnet=default \
--tags=http-server \
--metadata=startup-script='#! /bin/bash
 apt-get update
 apt-get install apache2 -y
 echo "Page served from: "Region 1"" | \
 tee /var/www/html/index.html
 systemctl restart apache2'
```

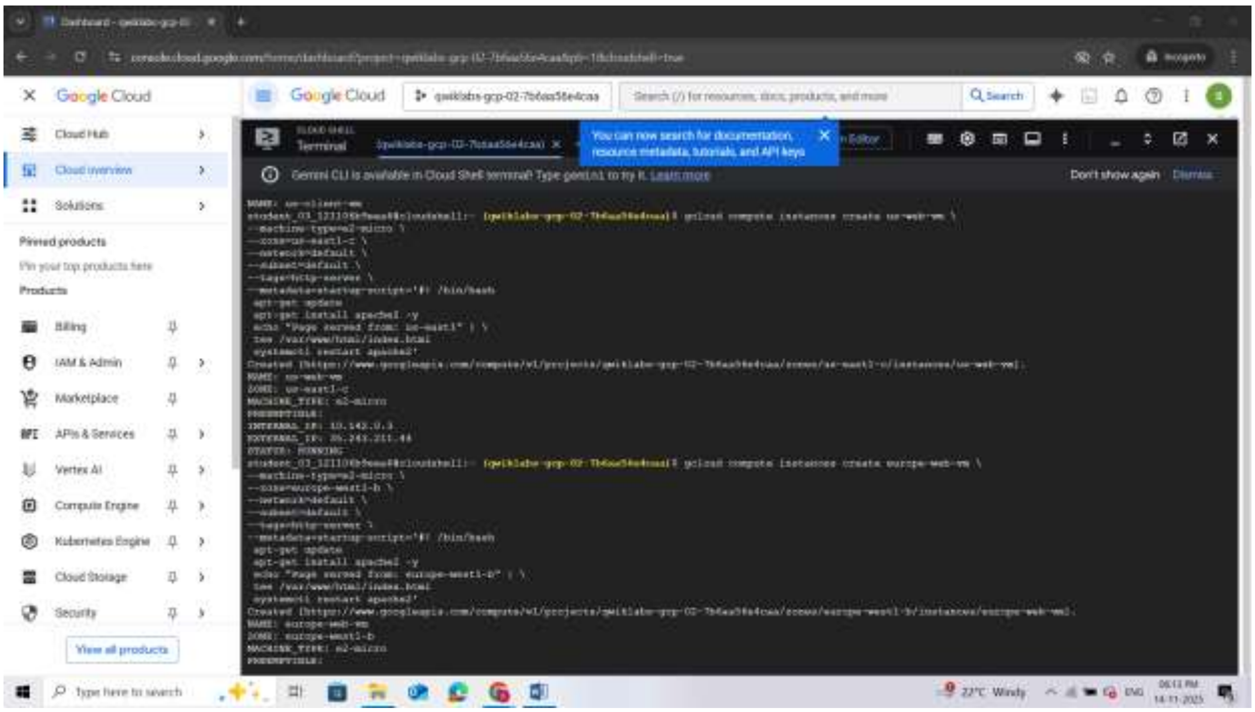
Note that you may see a different zone in gcloud than in the sample output shown below.

Launch server in Europe

- Run the following to command:

```
gcloud compute instances create europe-web-vm \
--machine-type=e2-micro \
--zone="Zone 2" \
--network=default \
--subnet=default \
--tags=http-server \
--metadata=startup-script='#! /bin/bash
 apt-get update
 apt-get install apache2 -y
 echo "Page served from: "Zone 2"" | \
 tee /var/www/html/index.html
 systemctl restart apache2'
```

Note that you may see a different zone in gcloud than in the sample output shown below.



Task 5. Setting up environment variables

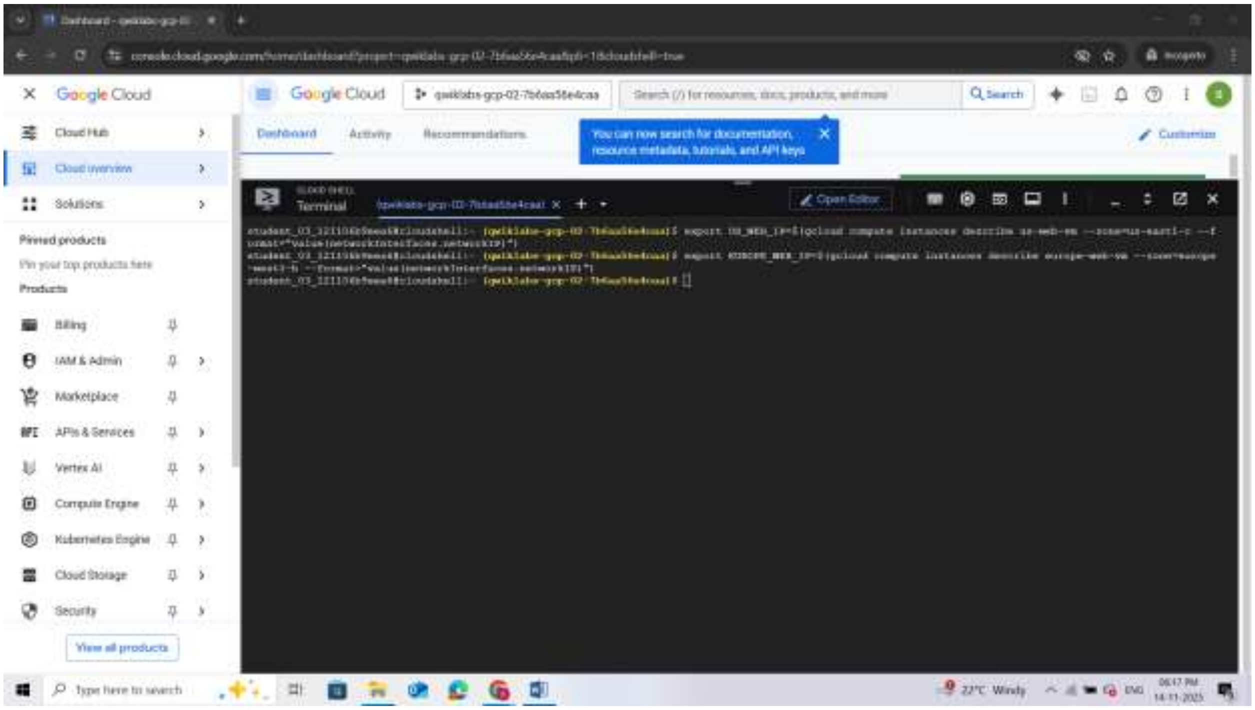
Before you configure Cloud DNS, note the Internal IP addresses of the web servers. You need these IPs to create the routing policy. In this section, you will use the gcloud compute instances describe command to save the internal IP addresses as environment variables.

1. Command to save IP address for the VM in the United States

export US_WEB_IP=$(gcloud compute instances describe us-web-vm --zone="Zone 1" --format="value(networkInterfaces.networkIP)")

2. Command to save the IP address for the VM in Europe:

export EUROPE_WEB_IP=$(gcloud compute instances describe europe-web-vm --zone="Zone 2" --format="value(networkInterfaces.networkIP)")



Task 6. Create the private zone

Now that your client and server VMs are running, it's time to configure the DNS settings. Before creating the A records for the web servers, you need to create the Cloud DNS Private Zone.

For this lab, use the example.com domain name for the Cloud DNS zone.

- Use the gcloud dns managed-zones create command to create the zone:
  gcloud dns managed-zones create example --description=test --dns-name=example.com --
  networks=default --visibility=private

Task 7. Create Cloud DNS Routing Policy

In this section, configure the Cloud DNS Geolocation Routing Policy. You will create a record set in the example.com zone that you created in the previous section.

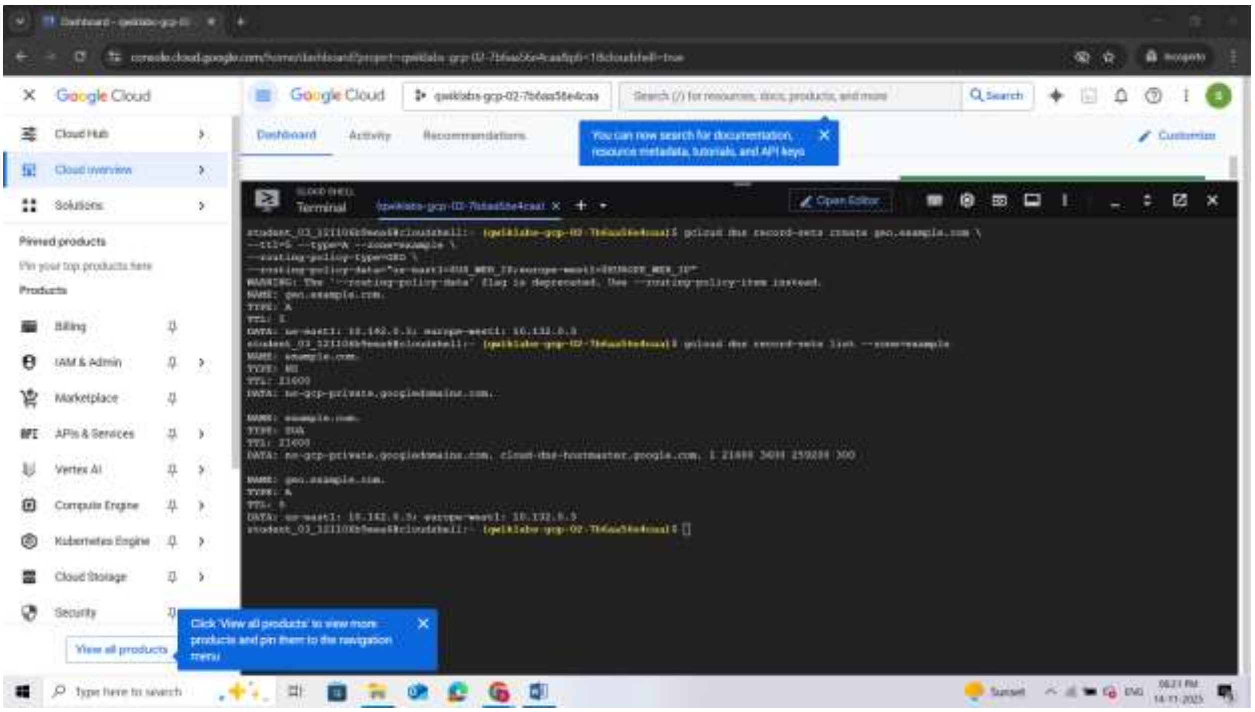
Create

- Use the gcloud dns record-sets create command to create the geo.example.com recordset:
  gcloud dns record-sets create geo.example.com \
  --ttl=5 --type=A --zone=example \
  --routing-policy-type=GEO \
  --routing-policy-data=""Region 1"=$US_WEB_IP;"Region 2"=$EUROPE_WEB_IP"
  You are creating an A record with a Time to Live (TTL) of 5 seconds. The policy type is GEO, and the routing_policy_data field accepts a semicolon-delimited list of the format ${region}:${rrdata},${rrdata}.

Verify

- Use the dns record-sets list command to verify that the geo.example.com DNS record is configured as expected:
  gcloud dns record-sets list --zone=example
  The output shows that an A record with a TTL of 5 is created for geo.example.com, and the data matches our server set up in each region.

  Note that in gcloud, the DATA value under geo.example.com may include United States and Europe regions that differ from the sample output below.



Task 8. Testing

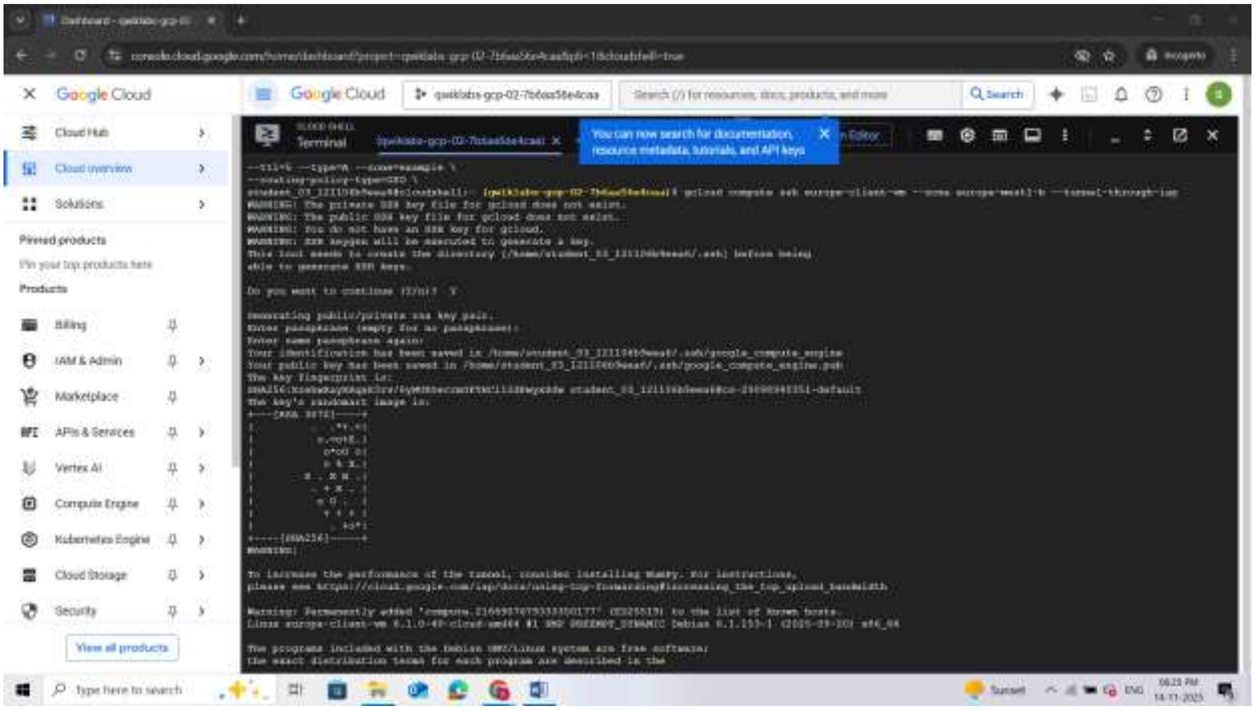
It's time to test the configuration. In this section, you will SSH into all the client VMs. Since all of the web server VMs are behind the geo.example.com domain, you will use CURL command to access this endpoint.

Since you are using a Geolocation policy, the expected result is that:

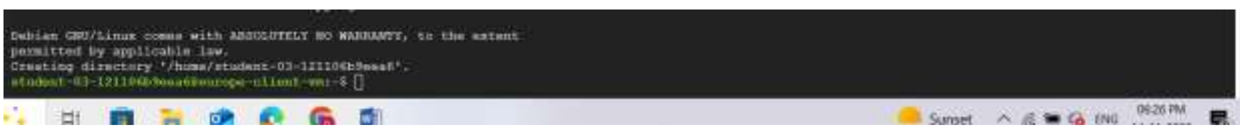
- The client in the US should always get a response from the <Zone 1 region.
- The client in Europe should always get a response from the Zone 2 region.

Testing from the client VM in Europe

1. Use the gcloud compute ssh command to log into the client VM:
gcloud compute ssh europe-client-vm --zone "Zone 2" --tunnel-through-iap
2. Follow prompts to SSH into the machine. When asked to enter the passphrase, leave the field blank and press the Enter key twice.



Once complete, the command line should change to "user_name@europe-client-vm:~$"

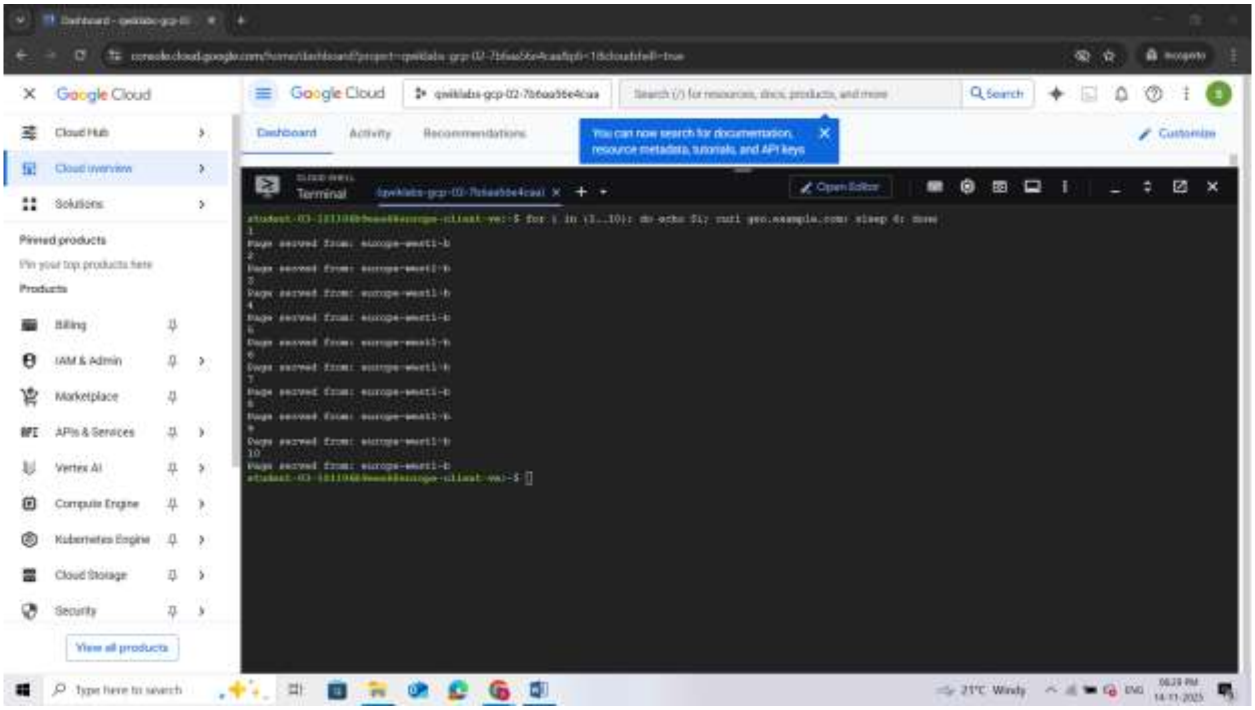


Use curl to access the web server

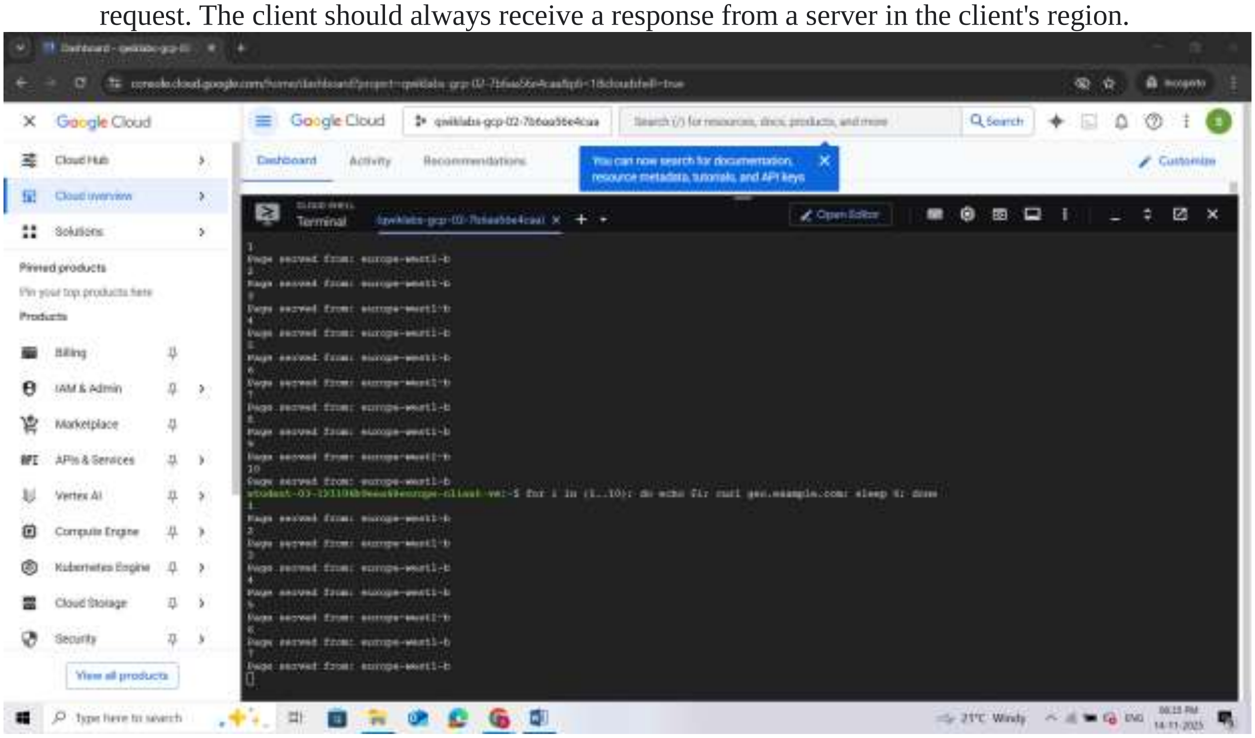
1. Now that you are in the client VM, use the CURL command to access the geo.example.com endpoint. The loop is configured to run the command ten times with a sleep timer of 6 seconds:

for i in {1..10}; do echo $i; curl geo.example.com; sleep 6; done

Since the TTL on the DNS record is set to 5 seconds, a sleep timer of 6 seconds has been added. The sleep timer will make sure that you get an uncached DNS response for each cURL request. This command will take approximately one minute to complete.

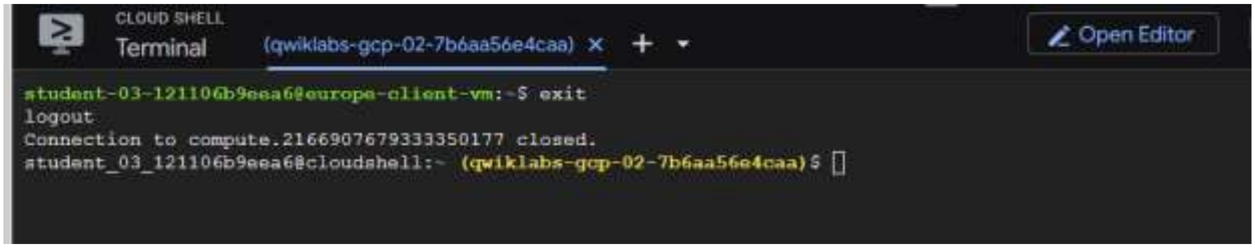


2. Run this test multiple times and analyze the output to see which server is responding to the request. The client should always receive a response from a server in the client's region.



Getting back to Cloud Shell

- Once you have run the test multiple times, exit the client VM in Europe by typing "exit" in the VM's command prompt. This will bring you back to the Cloud Shell console.



Testing from the client VM in us-east1

Now perform the same test from the client VM in the US.

1. Use the gcloud command below to SSH into the us-client-vm:

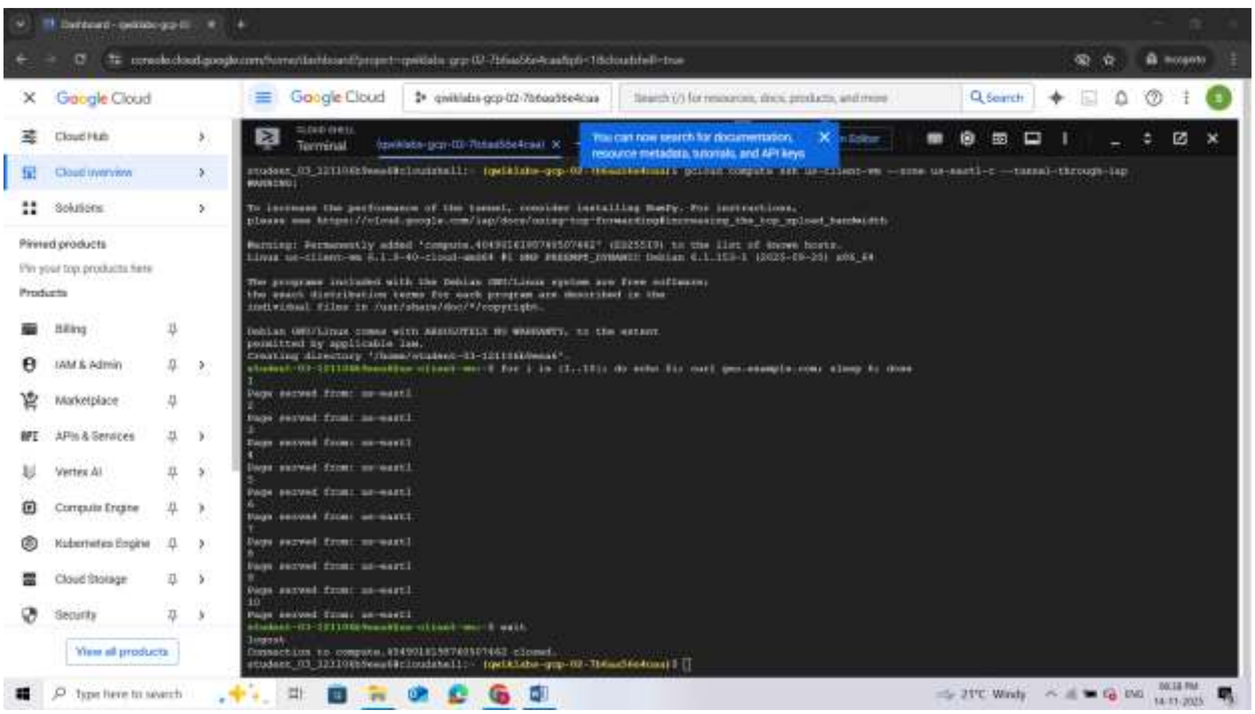
gcloud compute ssh us-client-vm --zone "Zone 1" --tunnel-through-iap

2. Use the curl command to access geo.example.com:

for i in {1..10}; do echo $i; curl geo.example.com; sleep 6; done

3. Now analyze the output to see which server is responding to the request. The client should always receive a response from a server in the client's region. The expected output is "Page served from: <filled at lab start>".

4. Once you have run the test multiple times, exit the client VM in the US by typing "exit" in the VM's command prompt.



Testing from the client VM in Asia

So far you have tested the setup from the United States and Europe. You have servers running in both the regions and have matching record sets for both the regions in Cloud DNS routing policy. There is no matching policy item for the region within Asia (selected earlier) in the Cloud DNS routing policy.

The Geolocation policy will apply a "nearest" match for source location when the source of the traffic doesn't match any policy items exactly. This means that the Asia client should be directed to the nearest web server.

In this section, you will resolve the geo.example.com domain from the client VM in Asia and will analyze the response.

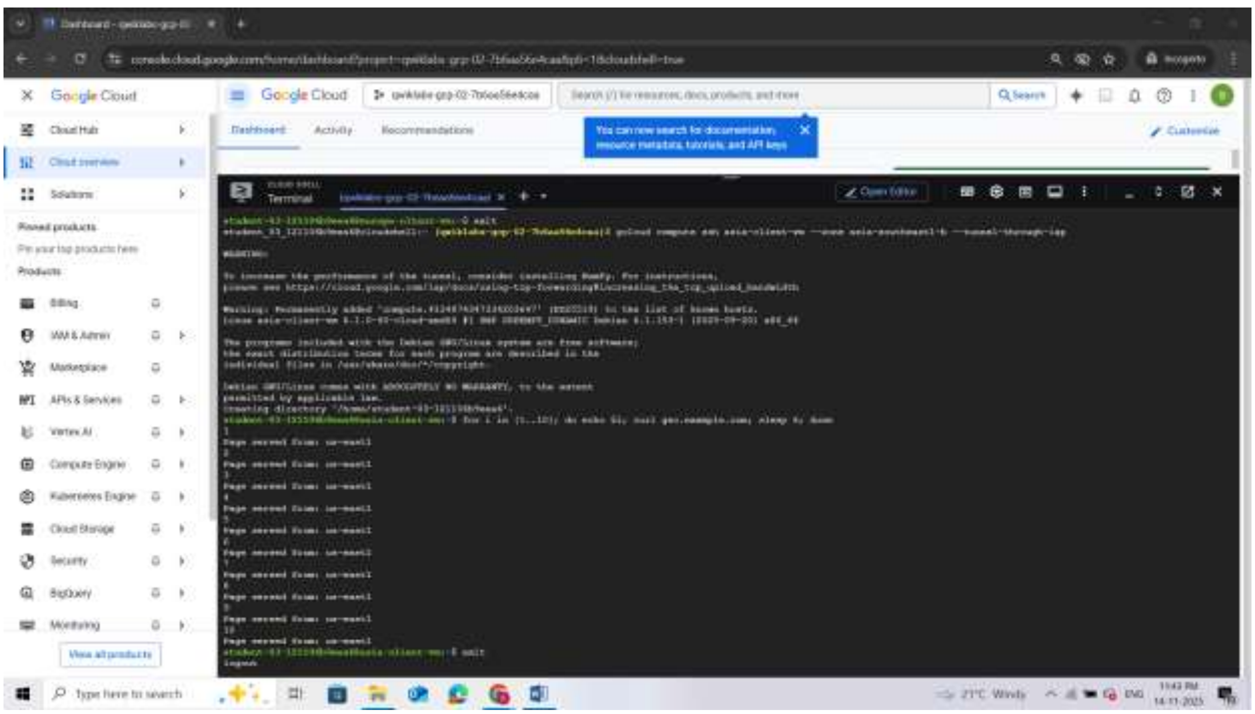
1. SSH into the asia-client-vm. For <SELECTED-ZONE>, use the zone that you used to create the Asia client.

gcloud compute ssh asia-client-vm --zone <SELECTED-ZONE> --tunnel-through-iap

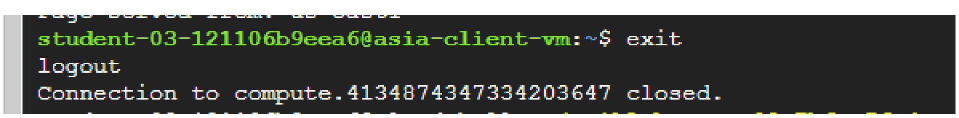
2. Then access geo.example.com:

for i in {1..10}; do echo $i; curl geo.example.com; sleep 6; done

3. Analyze the output to see which server is responding to the request. Since there is no policy item for any of the Asia regions, Cloud DNS will direct the client to the nearest server.



4. Once you have run the test multiple times, exit the client VM in Asia by typing "exit" in the VM's command prompt.



Task 9. Delete lab resources

Although all resources you used in this lab will be deleted when you finish, it is good practice to remove resources you no longer need to avoid unnecessary charges.

- The following gcloud commands will delete all the resources that were created in the lab. (Note that SELECTED-ZONE is the Asia zone that you wrote down earlier.)
  #delete VMS
  gcloud compute instances delete -q us-client-vm --zone "ZONE"

  gcloud compute instances delete -q us-web-vm --zone "ZONE"

  gcloud compute instances delete -q europe-client-vm --zone "Zone 2"

  gcloud compute instances delete -q europe-web-vm --zone "Zone 2"

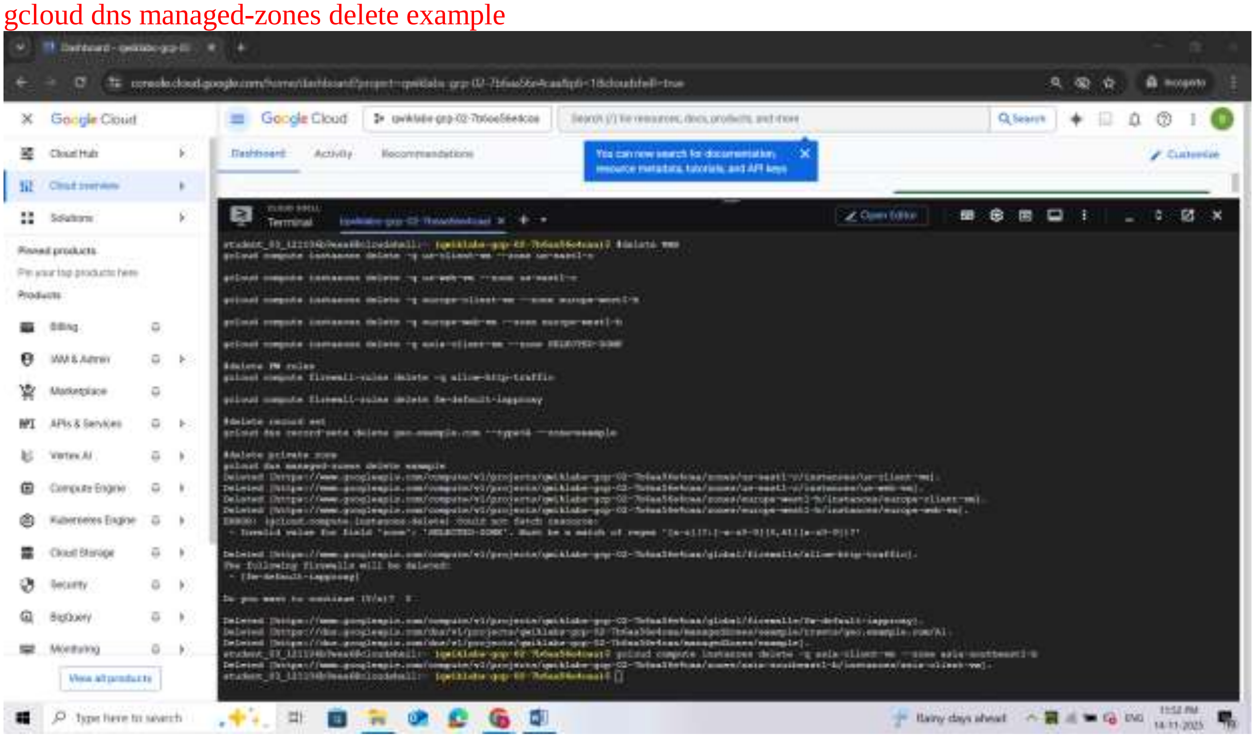
  gcloud compute instances delete -q asia-client-vm --zone SELECTED-ZONE

  #delete FW rules
  gcloud compute firewall-rules delete -q allow-http-traffic

  gcloud compute firewall-rules delete fw-default-iapproxy

  #delete record set
  gcloud dns record-sets delete geo.example.com --type=A --zone=example

  #delete private zone
  gcloud dns managed-zones delete example



Review

In this lab, you configured and used Cloud DNS routing policies with Geolocation routing policy. You also verified the configuration and behavior of the Cloud DNS routing policy by observing the HTTP response when accessing the web servers.

Next steps / Learn more

You can read more about Cloud DNS routing policies.

End your lab

When you have completed your lab, click **End Lab**. Google Cloud Skills Boost removes the resources you’ve used and cleans the account for you.

QUIZ

✓ 1. You must create a VM that has an IPv6 address. How do you do it?

- ✓ Create a dual-stack subnet, and create the VM with an IPv6 address.
- ○ Create a single-stack network, and create the VM with an IPv6 address.
- ○ Create an IPv6-only subnet, and create the VM with an IPv6 address.
- ○ Create a dual-stack network, and create the VM with an IPv6 address.

This is correct. Dual-stack subnets support both IPv4 and IPv6, allowing you to create VMs with both types of addresses.

✓ 2. To set up hybrid deployments for DNS resolution, which type of DNS policy should you use?

- ○ Response policy
- ○ Routing policy
- ○ Traffic policy
- ✓ Server policy

This is correct. Server policies are used to define which DNS servers should be queried for different DNS zones, making them suitable for hybrid deployments where you might have on-premises and cloud-based DNS servers.

LAB 2: Implement Private Google Access and Cloud NAT

Overview

In this lab, you implement Private Google Access and Cloud NAT for a VM instance that doesn't have an external IP address. Then, you verify access to public IP addresses of Google APIs and services and other connections to the internet.

VM instances without external IP addresses are isolated from external networks. Using Cloud NAT, these instances can access the internet for updates and patches, and in some cases, for **bootstrapping**. As a managed service, Cloud NAT provides high availability without user management and intervention.

Objectives

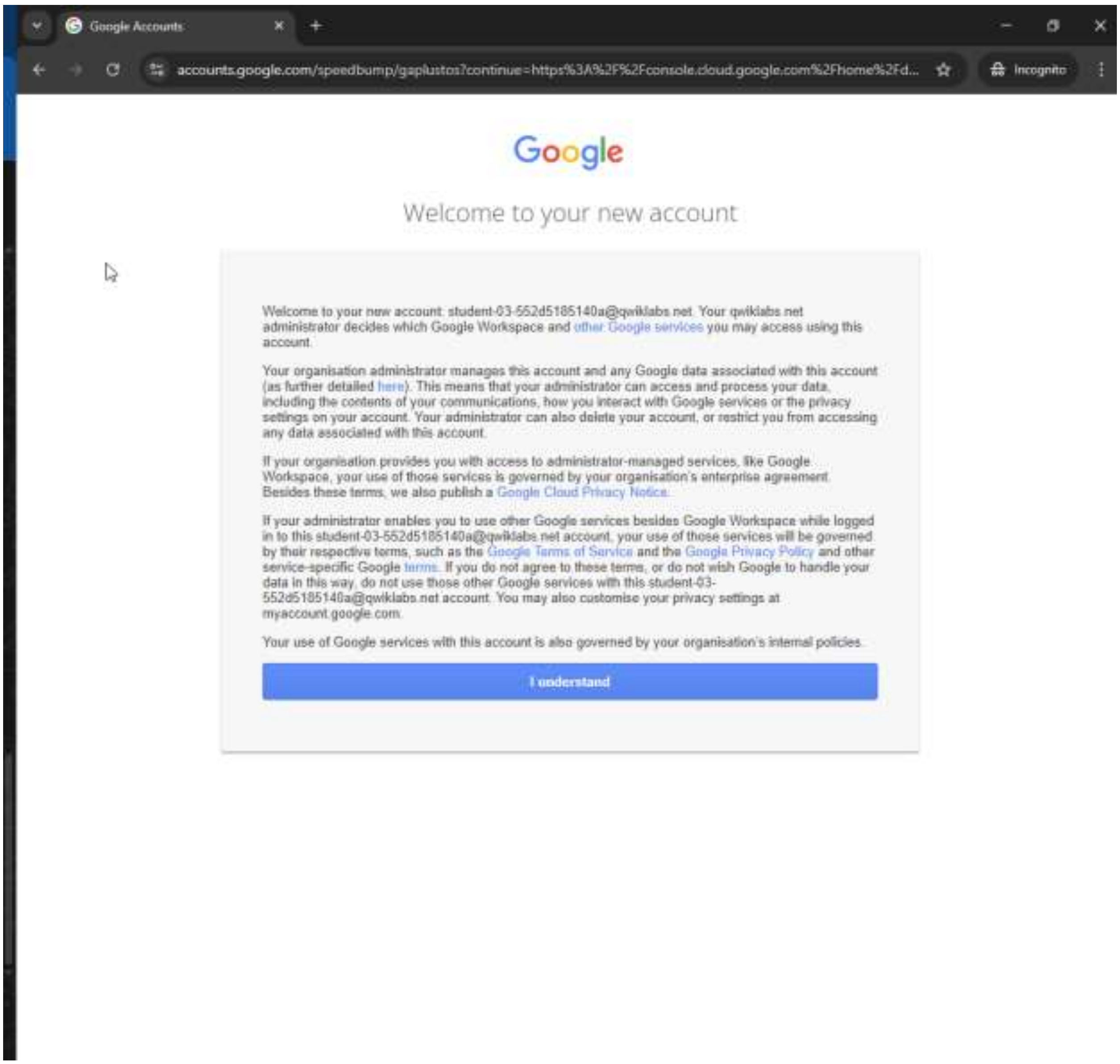
In this lab, you learn how to perform the following tasks:

- Configure a VM instance that doesn't have an external IP address
- Connect to a VM instance using an Identity-Aware Proxy (IAP) tunnel
- Enable Private Google Access on a subnet
- Configure a Cloud NAT gateway
- Verify access to public IP addresses of Google APIs and services and other connections to the internet

Setup and Requirements

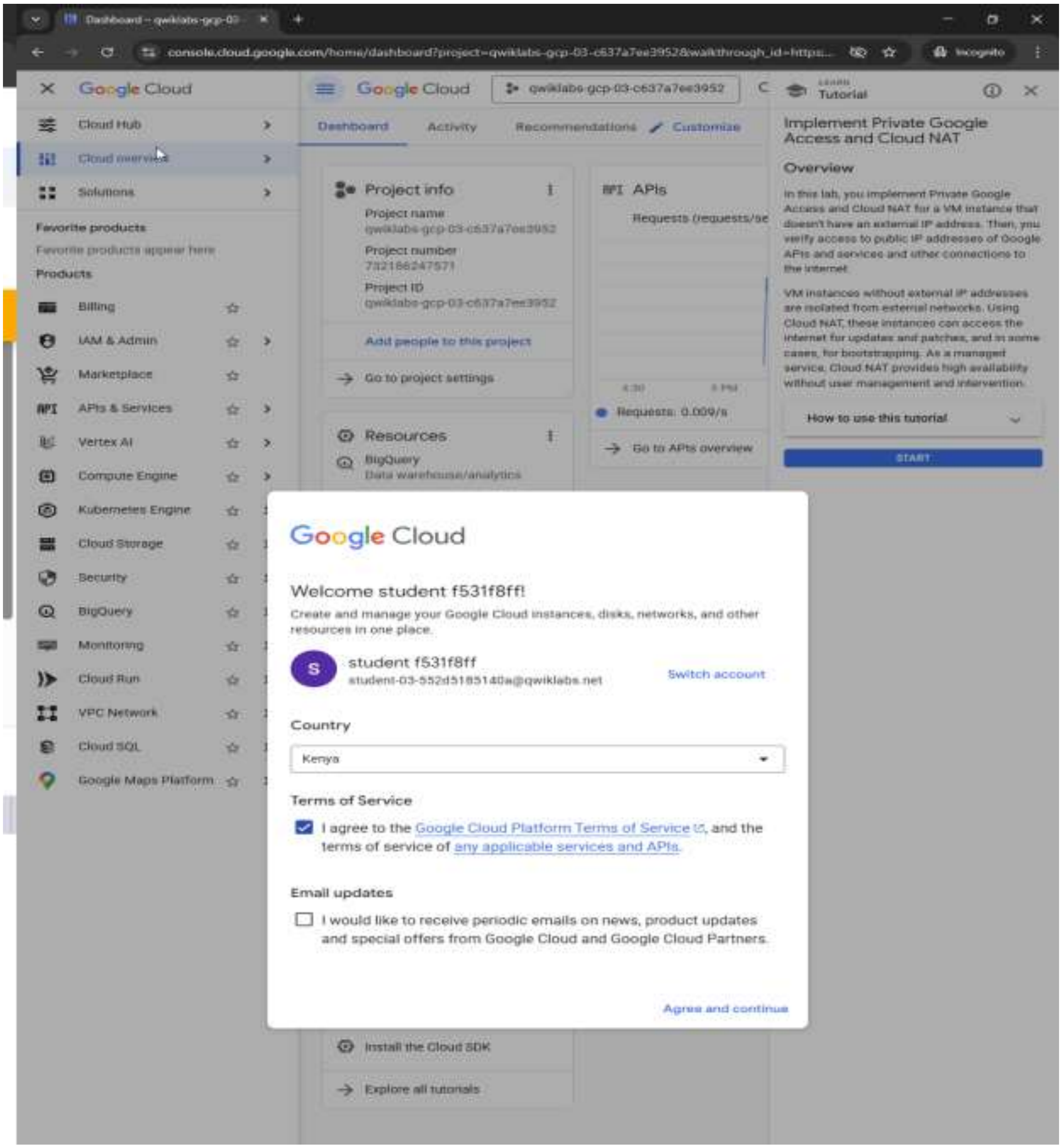
For each lab, you get a new Google Cloud project and set of resources for a fixed time at no cost.

1. Click the **Start Lab** button. If you need to pay for the lab, a pop-up opens for you to select your payment method. On the left is the **Lab Details** panel with the following:
    - The **Open Google Cloud console** button
    - Time remaining
    - The temporary credentials that you must use for this lab
    - Other information, if needed, to step through this lab
2. Click **Open Google Cloud console** (or right-click and select **Open Link in Incognito Window** if you are running the Chrome browser).



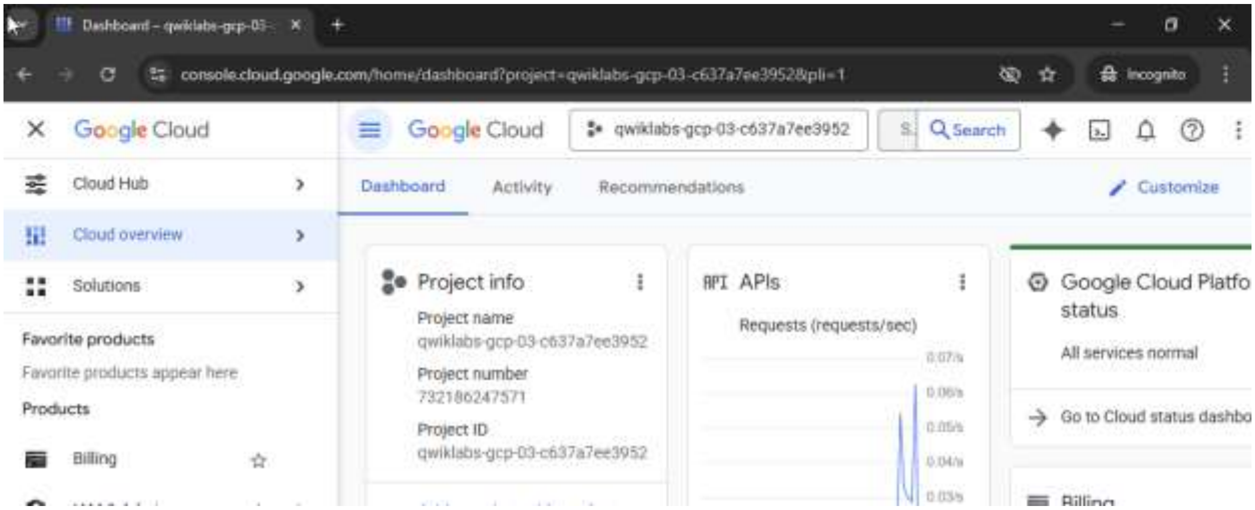
The lab spins up resources, and then opens another tab that shows the **Sign in** page.

3. If necessary, copy the **Username** below and paste it into the **Sign in** dialog. You can also find the **Username** in the **Lab Details** panel.
4. Click Next.
5. Copy the Password below and paste it into the Welcome dialog. You can also find the **Password** in the **Lab Details** panel.
6. Click Next.
7. Click through the subsequent pages:
   - Accept the terms and conditions.



- Do not add recovery options or two-factor authentication (because this is a temporary account).
- Do not sign up for free trials.

After a few moments, the Google Cloud console opens in this tab.



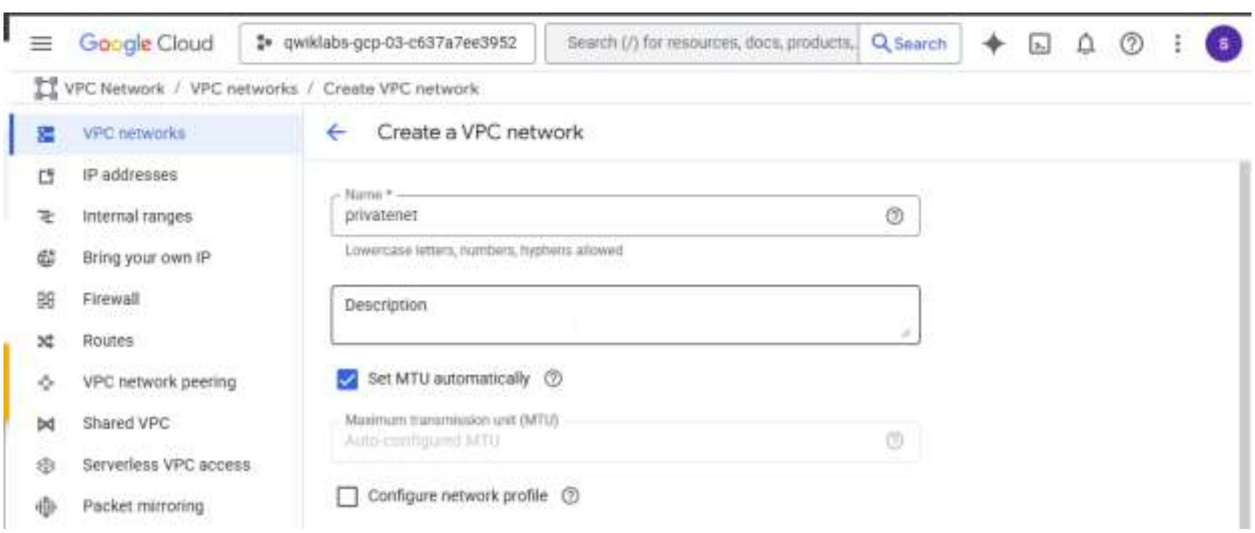
Task 1. Create the VM instance

Create a VPC network with some firewall rules and a VM instance that has no external IP address, and connect to the instance using an IAP tunnel.

Create a VPC network and firewall rules

First, create a VPC network for the VM instance and a firewall rule to allow SSH access.

1. In the Google Cloud console, in the **Navigation menu** (☰), click **VPC network > VPC networks**.
2. Click **Create VPC Network**.
3. For **Name**, type **privatenet**.

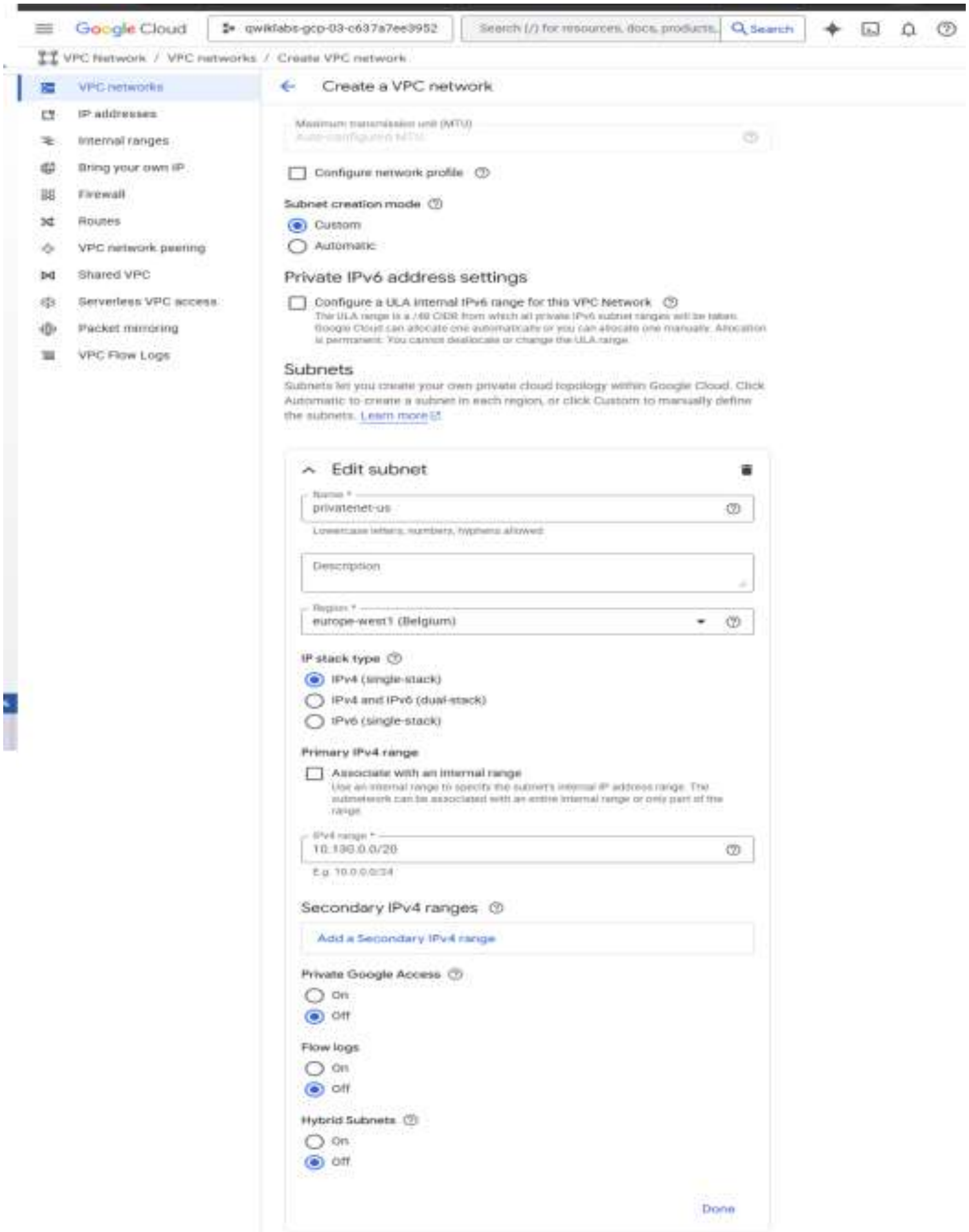


4. For **Subnet creation mode**, click **Custom**.
5. In **New Subnet** specify the following, and leave the remaining settings as their defaults:

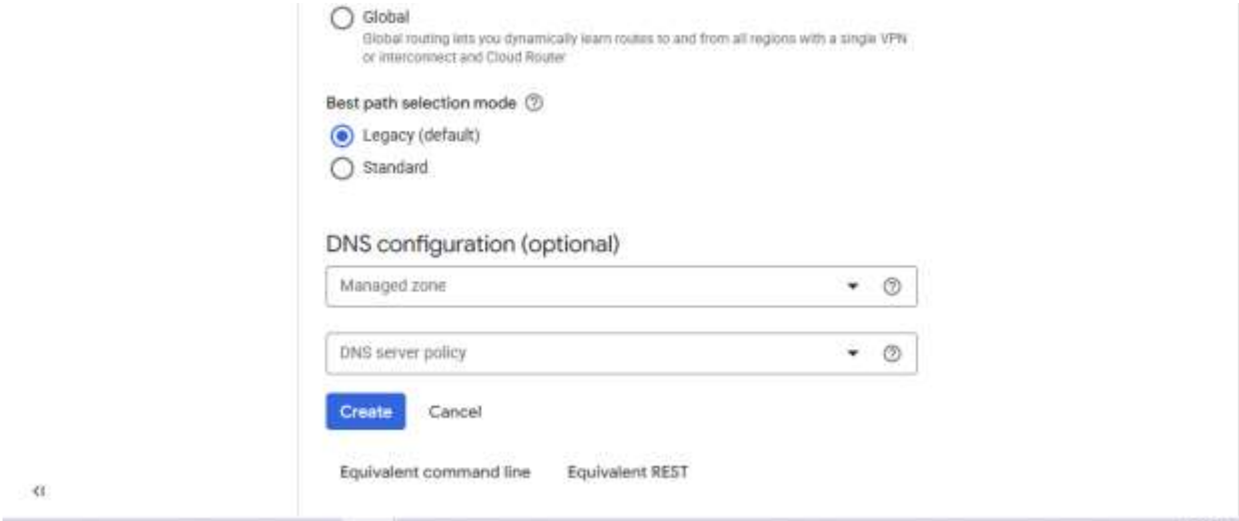
Property	**Value (type value or select option as specified)**
Name	privatenet-us
Region	REGION
IPv4 range	10.130.0.0/20

Note: *Don't enable* ***Private Google access*** *yet!*

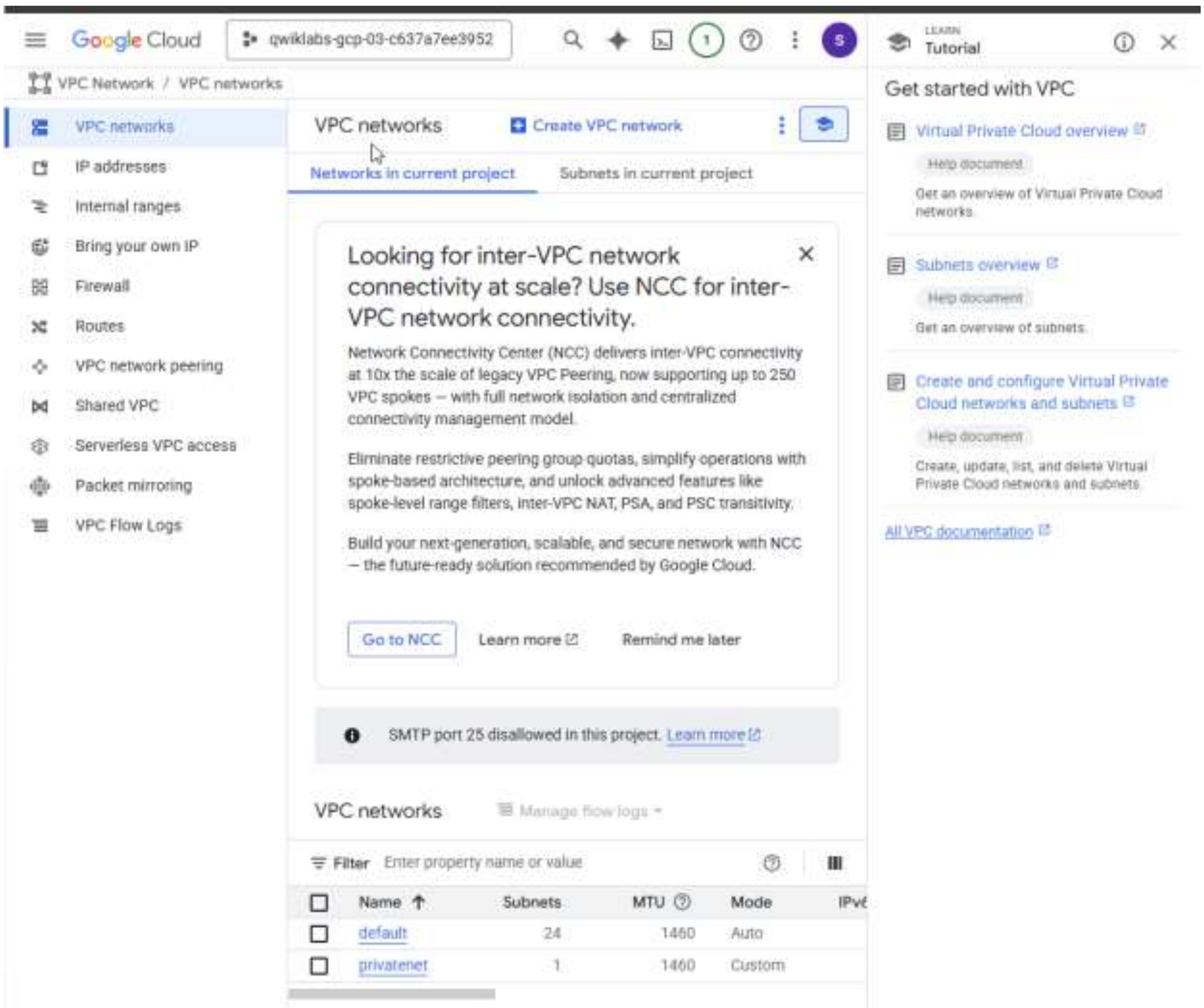
6. Click **Done**.



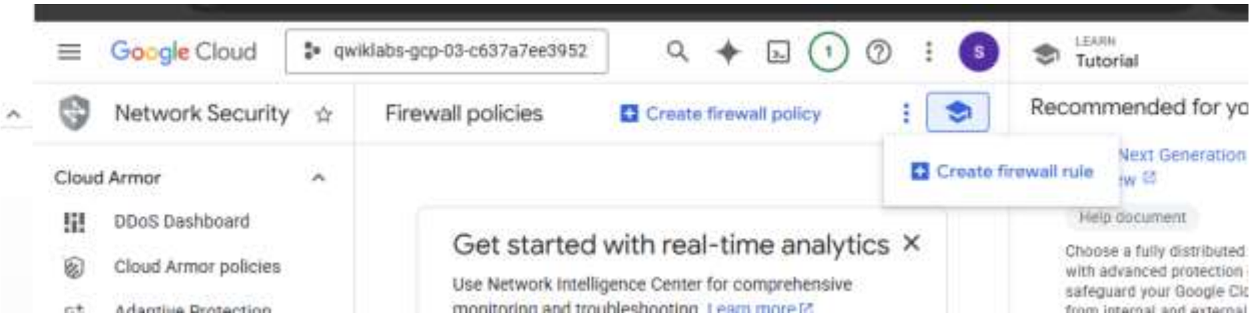
7. Click **Create** and wait for the network to be created.



8. In the left pane, click **Firewall**.



9. Click **Create Firewall Rule**.

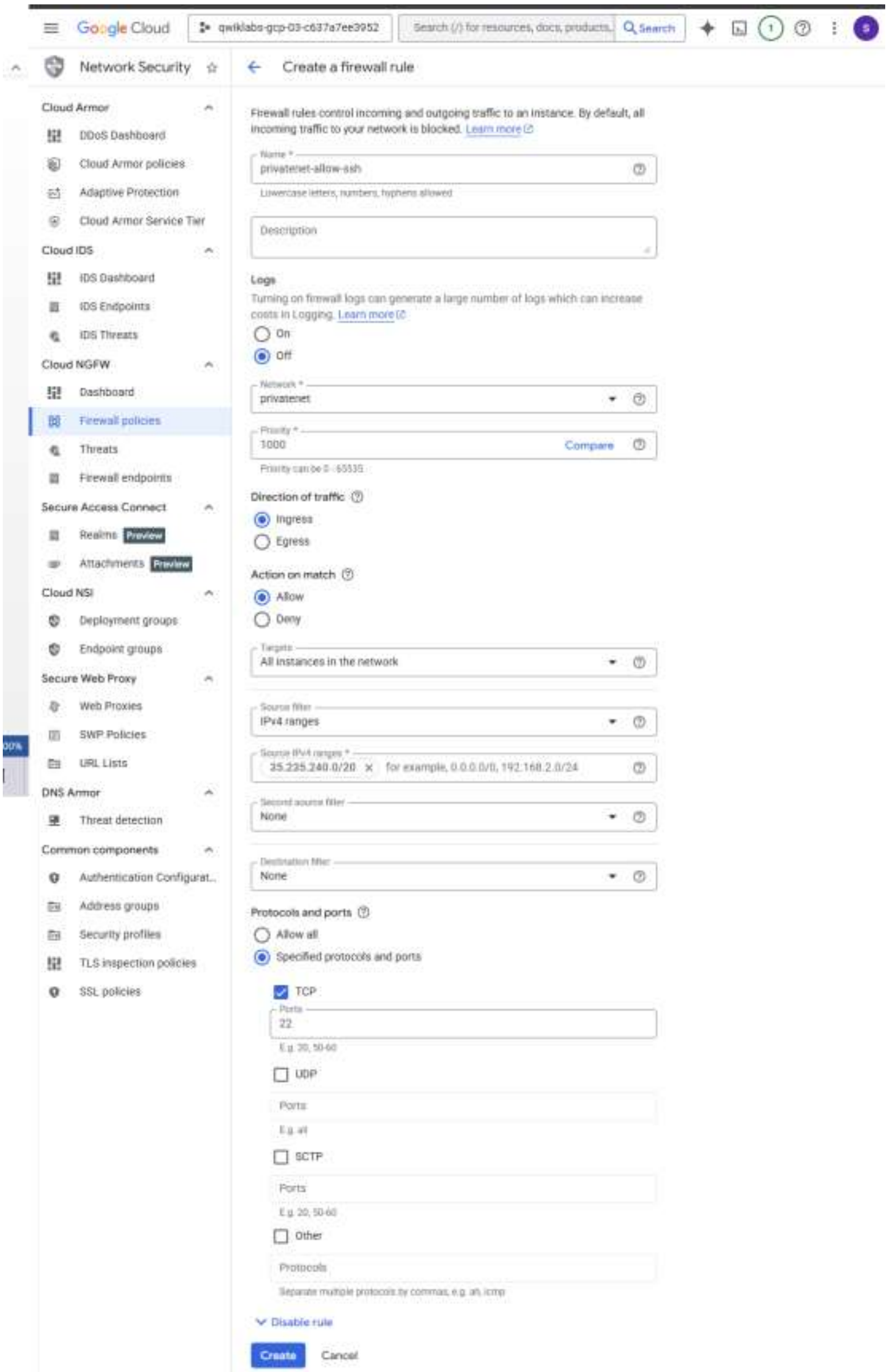


10. Specify the following, and leave the remaining settings as their defaults:

Property	Value (type value or select option as specified)
Name	privatenet-allow-ssh
Network	privatenet
Targets	All instances in the network
Source filter	IPv4 ranges
Source IPv4 ranges	35.235.240.0/20
Protocols and ports	Specified protocols and ports

11. For **tcp**, click the checkbox and specify port **22**.

12. Click **Create**.



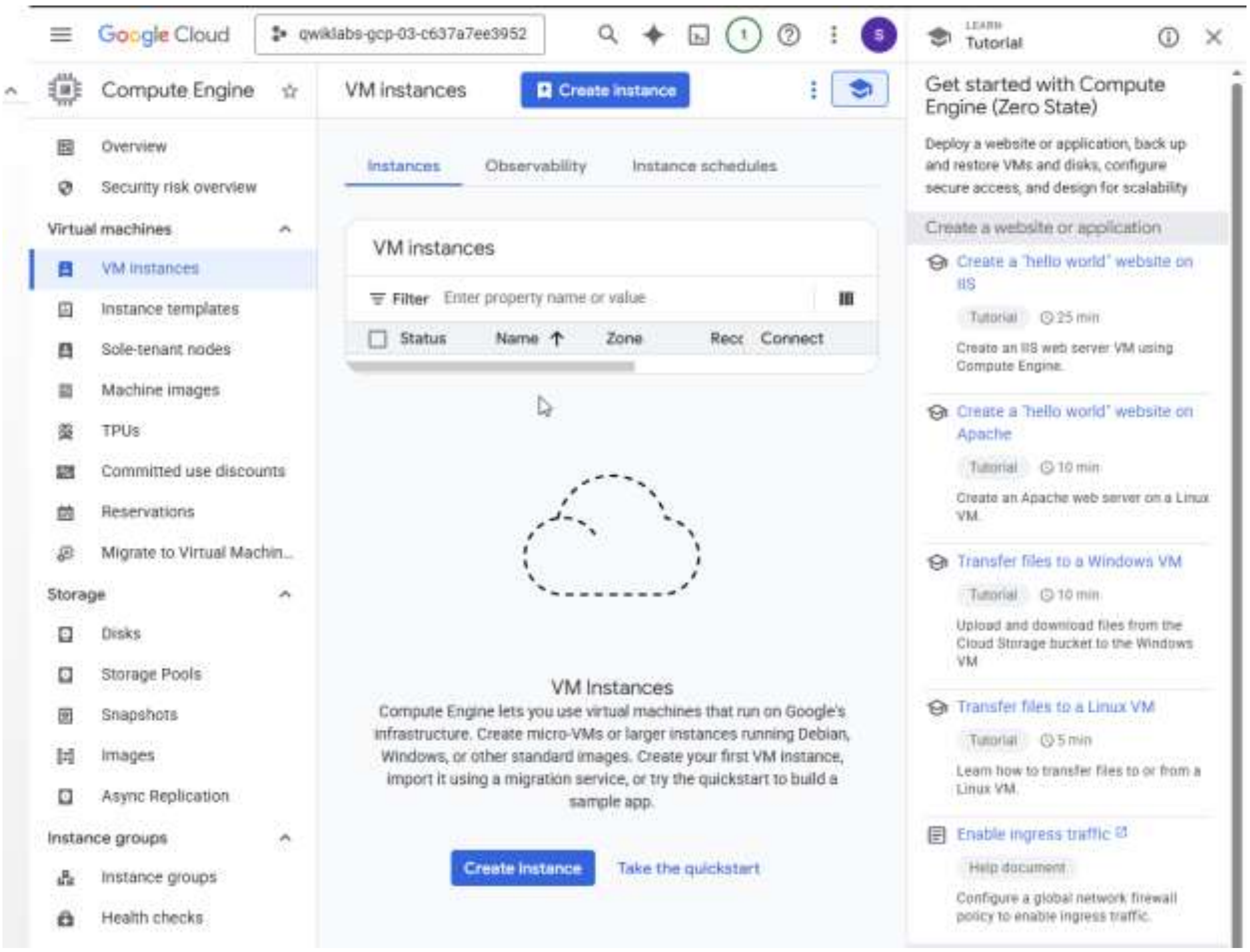
Google Cloud
qwiklabs-gcp-03-c637a7ee3952
Search (/) for resources, docs, products,
Search
Network Security
Create a firewall rule
Cloud Armor
DDoS Dashboard
Cloud Armor policies
Adaptive Protection
Cloud Armor Service Tier
Cloud IDS
IDS Dashboard
IDS Endpoints
IDS Threats
Cloud NGFW
Dashboard
Firewall policies
Threats
Firewall endpoints
Secure Access Connect
Realms
Preview
Attachments
Preview
Cloud NSI
Deployment groups
Endpoint groups
Secure Web Proxy
Web Proxies
SWP Policies
URL Lists
DNS Armor
Threat detection
Common components
Authentication Configurat...
Address groups
Security profiles
TLS inspection policies
SSL policies
Firewall rules control incoming and outgoing traffic to an instance. By default, all incoming traffic to your network is blocked. Learn more
Name *
privatenet-allow-ssh
Lowercase letters, numbers, hyphens allowed
Description
Logs
Turning on firewall logs can generate a large number of logs which can increase costs in Logging. Learn more
On
Off
Network *
privatenet
Priority *
1000
Compare
Priority can be 0 - 65535
Direction of traffic
Ingress
Egress
Action on match
Allow
Deny
Targets
All instances in the network
Source filter
IPv4 ranges
Source IPv4 ranges *
35.235.240.0/20
for example, 0.0.0.0/0, 192.168.2.0/24
Second source filter
None
Destination filter
None
Protocols and ports
Allow all
Specified protocols and ports
TCP
Ports
22
E.g. 20, 50-60
UDP
Ports
E.g. all
SCTP
Ports
E.g. 20, 50-60
Other
Protocols
Separate multiple protocols by commas, e.g. ah, icmp
Disable rule
Create
Cancel

Note: *In order to connect to your private instance using SSH, you need to open an appropriate port on the firewall. [IAP connections](#) come from a specific set of IP addresses (**35.235.240.0/20**). Therefore, you can limit the rule to this CIDR range.*

Create the VM instance with no public IP address

1. In the Google Cloud console, in the **Navigation menu** (☰), click **Compute Engine > VM instances**.

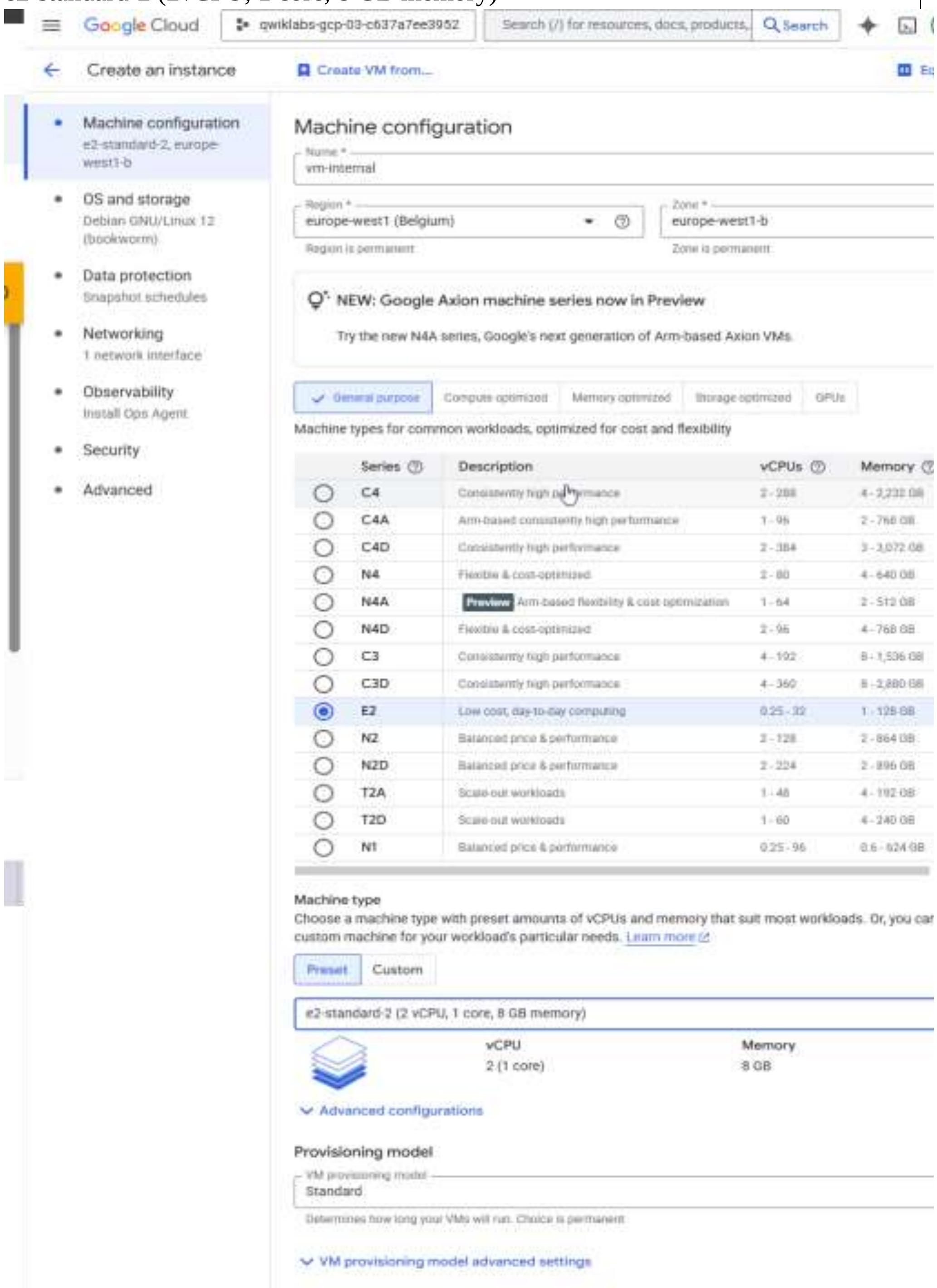
2. Click **Create Instance**.



3. On the **Machine configuration** page, specify the following, and leave the remaining settings as their defaults:

Property	Value (type value or select option as specified)
Name	vm-internal

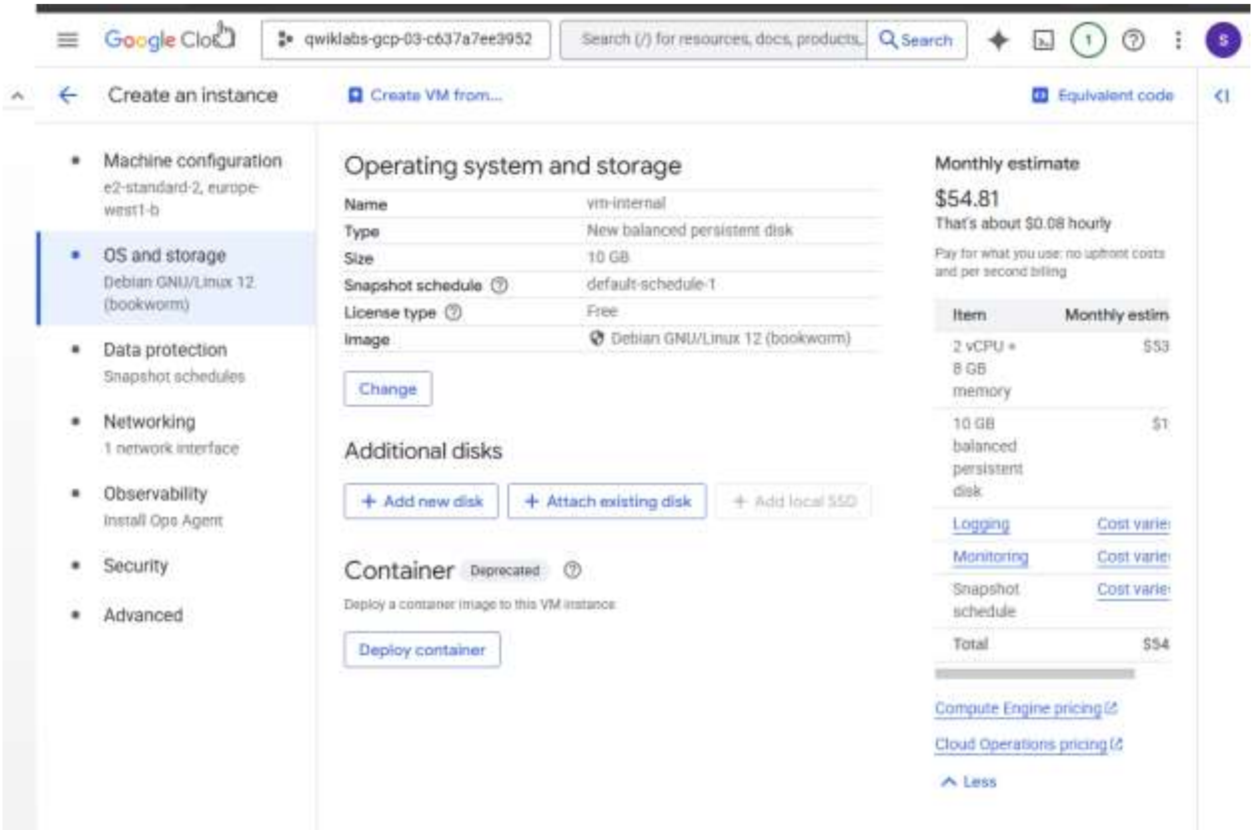
Region	europe-west1
Zone	europe-west1-b
Series	E2

<table>
<tr>
<td>Machine type</td>
<td>e2-standard-2 (2vCPU, 1 core, 8 GB memory)


</td>
</tr>
</table>

4. Click **OS and storage**.

5. If the **Image** shown is not **Debian GNU/Linux 12 (bookworm)**, click **Change** and select **Debian GNU/Linux 12 (bookworm)**, and then click **Select**.



6. Click **Networking**.

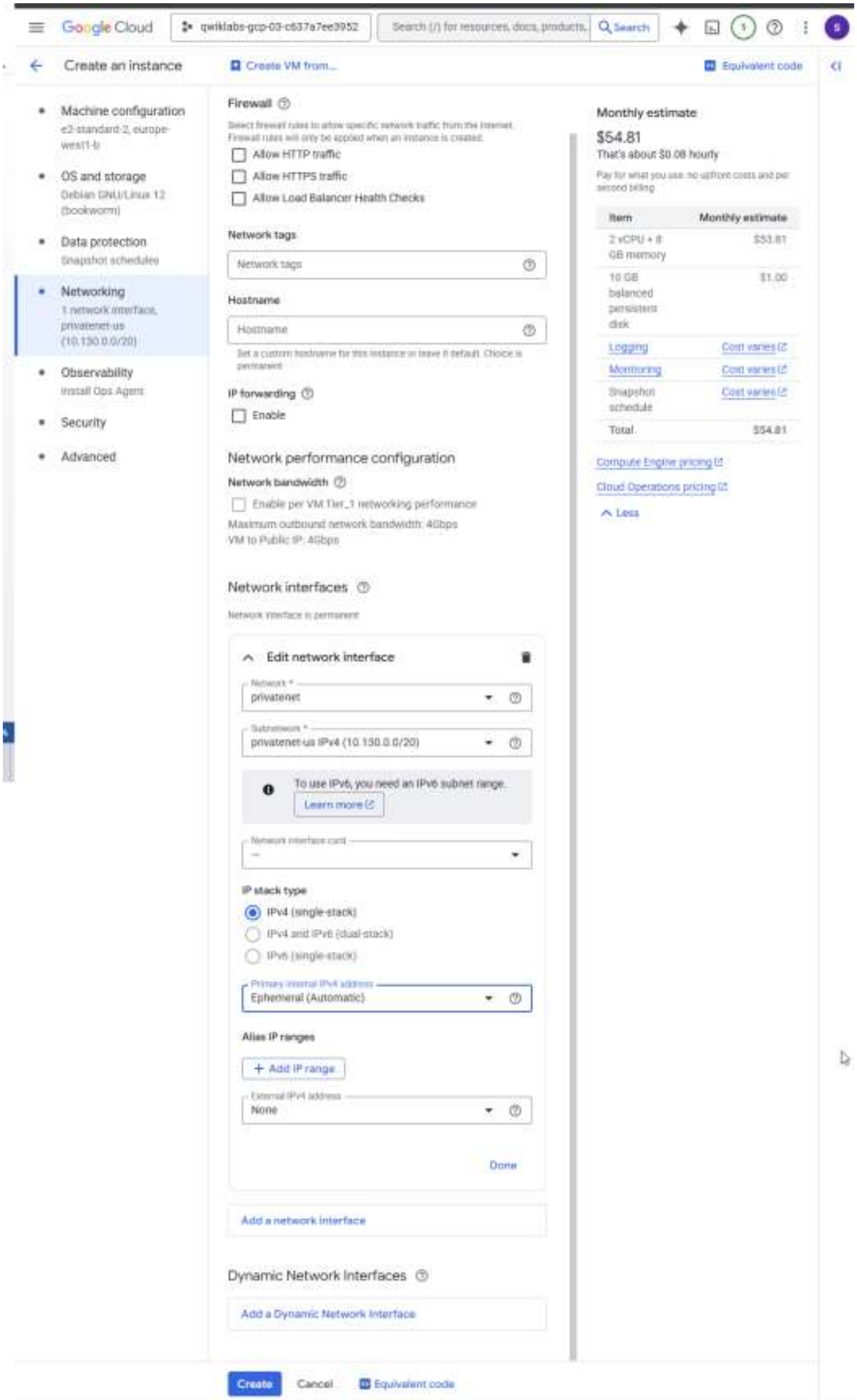
7. In **Network interfaces**, edit the network interface by specifying the following:

Property	Value (type value or select option as specified)
Network	privatenet
Subnetwork	privatenet-us
External IPv4 address	None

Note: *The default setting for a VM instance is to have an ephemeral external IP address. This behavior can be changed with a policy constraint at the organization or project level.*

8. Click **Done**.

9. Click **Create**, and wait for the VM instance to be created.

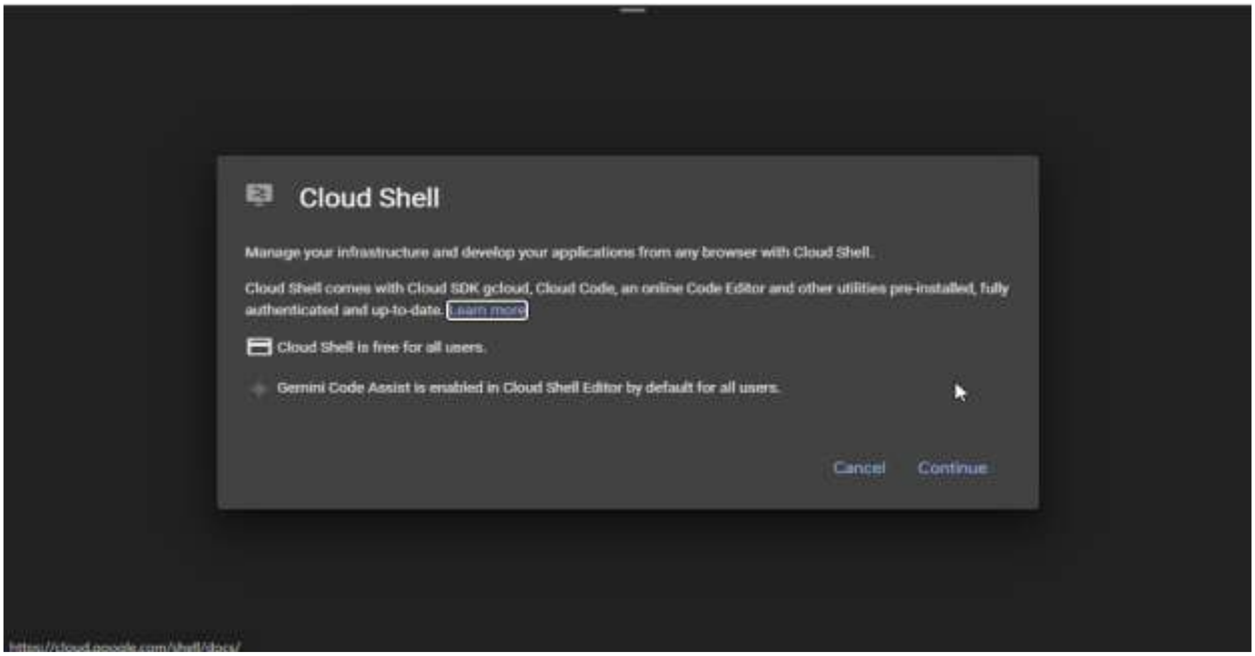

Google Cloud
qwiklabs-gcp-03-c637a7ee3952
Search (/) for resources, docs, products,
Search
Create an instance
Create VM from...
Equivalent code
Machine configuration
e2-standard-2, europe-west1-b
OS and storage
Debian GNU/Linux 12 (bookworm)
Data protection
Snapshot schedules
Networking
1 network interface, privatenet-us (10.130.0.0/20)
Observability
Install Ops Agent
Security
Advanced
Firewall
Allow HTTP traffic
Allow HTTPS traffic
Allow Load Balancer Health Checks
Network tags
Hostname
IP forwarding
Enable
Network performance configuration
Network bandwidth
Enable per VM Tier_1 networking performance
Maximum outbound network bandwidth: 4Gbps
VM to Public IP: 4Gbps
Network interfaces
Network interface is permanent
Edit network interface
Network *
privatenet
Subnetwork *
privatenet-us IPv4 (10.130.0.0/20)
To use IPv6, you need an IPv6 subnet range.
Learn more
Network interface card
IP stack type
IPv4 (single-stack)
IPv4 and IPv6 (dual-stack)
IPv6 (single-stack)
Primary internal IPv4 address
Ephemeral (Automatic)
Alias IP ranges
Add IP range
External IPv4 address
None
Done
Add a network interface
Dynamic Network Interfaces
Add a Dynamic Network Interface
Monthly estimate
$54.81
That's about $0.08 hourly
Pay for what you use: no upfront costs and per second billing
Item
Monthly estimate
2 vCPU + 8 GB memory
$53.81
10 GB balanced persistent disk
$1.00
Logging
Cost varies
Monitoring
Cost varies
Snapshot schedule
Cost varies
Total
$54.81
Compute Engine pricing
Cloud Operations pricing
Less
Create
Cancel
Equivalent code

10. On the **VM instances** page, verify that the **External IP** of **vm-internal** is **None**

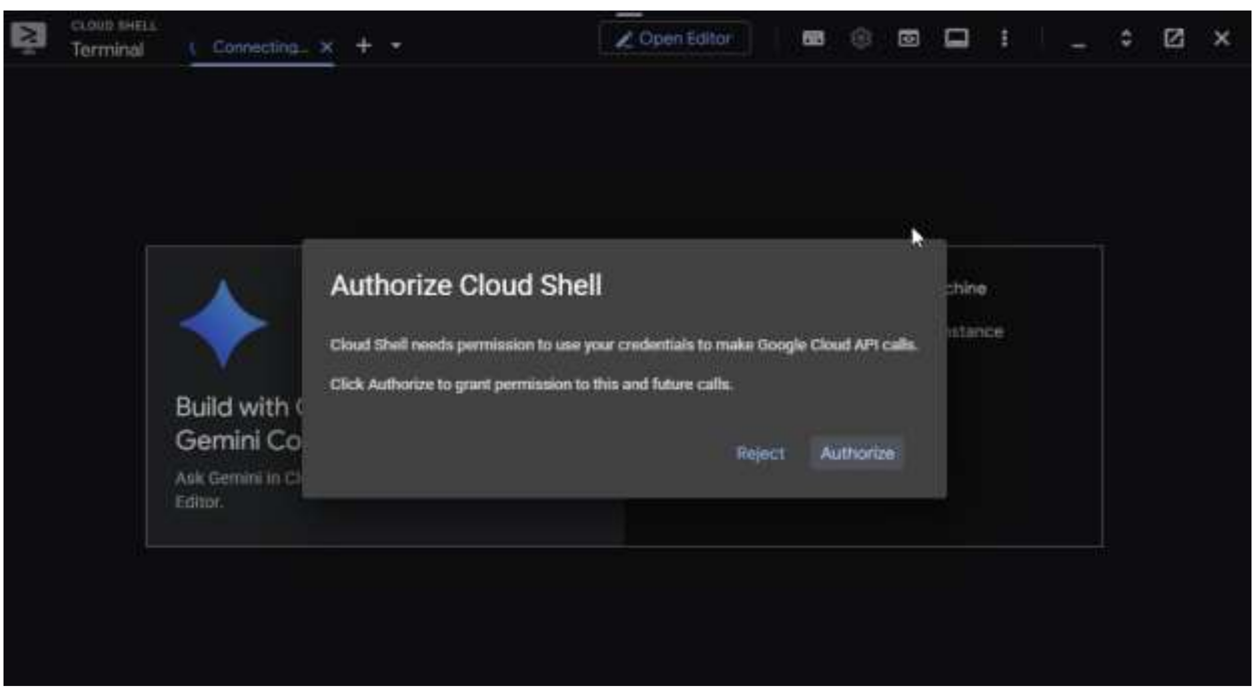
SSH to vm-internal to test the IAP tunnel

1. In the Cloud console, click **Activate Cloud Shell** (>_).

2. If prompted, click **Continue**.



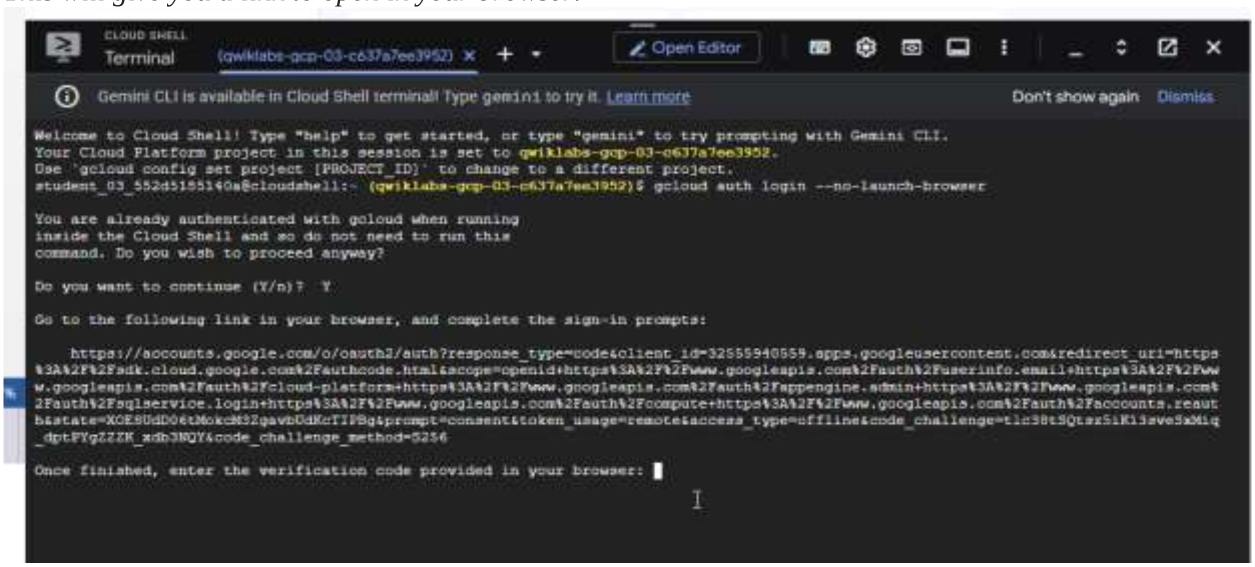
3. If prompted to authorize, click **Authorize**.



4. Run the following command in Cloud Shell to setup **auth** without opening up a browser: **gcloud auth login --no-launch-browser**

If prompted [Y/n], press **Y** and then **ENTER**.

This will give you a link to open in your browser.



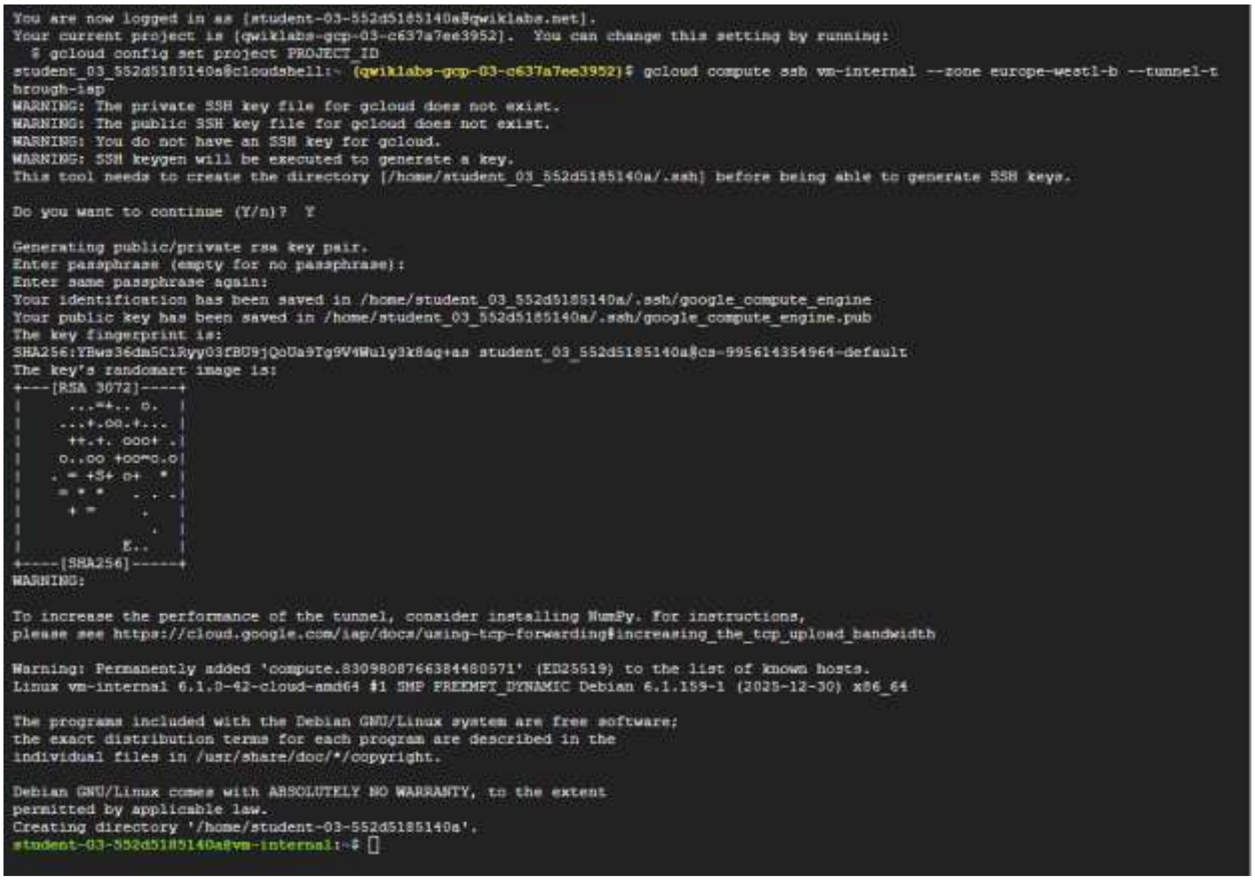
Open the link in the same browser where you are logged in to the lab account. Once you login you will get a verification code to copy. Paste that code in the cloud shell.

5. To connect to **vm-internal**, run the following command:
6. If prompted to continue, type **Y**.
7. When prompted for a passphrase, press **ENTER**.
8. When prompted for the same passphrase, press **ENTER**.

Did the command prompt change to @vm-internal?

True

False



```
You are now logged in as [student-03-552d5185140a@qwiklabs.net].
Your current project is [qwiklabs-gcp-03-c637a7ee3952].  You can change this setting by running:
  $ gcloud config set project PROJECT_ID
student_03_552d5185140a@cloudshell:~ (qwiklabs-gcp-03-c637a7ee3952)$ gcloud compute ssh vm-internal --zone europe-west1-b --tunnel-t
hrough-iap
WARNING: The private SSH key file for gcloud does not exist.
WARNING: The public SSH key file for gcloud does not exist.
WARNING: You do not have an SSH key for gcloud.
WARNING: SSH keygen will be executed to generate a key.
This tool needs to create the directory [/home/student_03_552d5185140a/.ssh] before being able to generate SSH keys.

Do you want to continue (Y/n)?  Y

Generating public/private rsa key pair.
Enter passphrase (empty for no passphrase):
Enter same passphrase again:
Your identification has been saved in /home/student_03_552d5185140a/.ssh/google_compute_engine
Your public key has been saved in /home/student_03_552d5185140a/.ssh/google_compute_engine.pub
The key fingerprint is:
SHA256:YBws36dm5CiRyy03fBU9jQoUa9Tg9V4Wuly3k8ag+as student_03_552d5185140a@cs-995614354964-default
The key's randomart image is:
+---[RSA 3072]----+
|     ...=+.. o.  |
|    ...+.oo.+... |
|     ++.+. ooo+ .|
|    o..oo +oo=o.o|
|   . = +S+ o+  * |
|    = * *    . . -|
|     + =     .   |
|              .  |
|          E..    |
+----[SHA256]-----+
WARNING:

To increase the performance of the tunnel, consider installing NumPy. For instructions,
please see https://cloud.google.com/iap/docs/using-tcp-forwarding#increasing_the_tcp_upload_bandwidth

Warning: Permanently added 'compute.8309808766384480571' (ED25519) to the list of known hosts.
Linux vm-internal 6.1.0-42-cloud-amd64 #1 SMP PREEMPT_DYNAMIC Debian 6.1.159-1 (2025-12-30) x86_64

The programs included with the Debian GNU/Linux system are free software;
the exact distribution terms for each program are described in the
individual files in /usr/share/doc/*/copyright.

Debian GNU/Linux comes with ABSOLUTELY NO WARRANTY, to the extent
permitted by applicable law.
Creating directory '/home/student-03-552d5185140a'.
student-03-552d5185140a@vm-internal:~$
```

9. To test the external connectivity of **vm-internal**, run the following command:

*This should not work because **vm-internal** has no external IP address!*

10. Wait for the **ping** command to complete.
11. To return to your Cloud Shell instance, run the following command:

```
student-03-552d5185140a@vm-internal:~$ ping -c 2 www.google.com
PING www.google.com (172.253.120.104) 56(84) bytes of data.

--- www.google.com ping statistics ---
2 packets transmitted, 0 received, 100% packet loss, time 1014ms

student-03-552d5185140a@vm-internal:~$ exit
logout
Connection to compute.8309808766384480571 closed.
student_03_552d5185140a@cloudshell:~ (qwiklabs-gcp-03-c637a7ee3952)$
student_03_552d5185140a@cloudshell:~ (qwiklabs-gcp-03-c637a7ee3952)$
student_03_552d5185140a@cloudshell:~ (qwiklabs-gcp-03-c637a7ee3952)$
```

Note: When instances do not have external IP addresses, they can only be reached by other instances on the network via a managed VPN gateway or via a Cloud IAP tunnel. Cloud IAP enables context-aware access to VMs via SSH and RDP without bastion hosts.

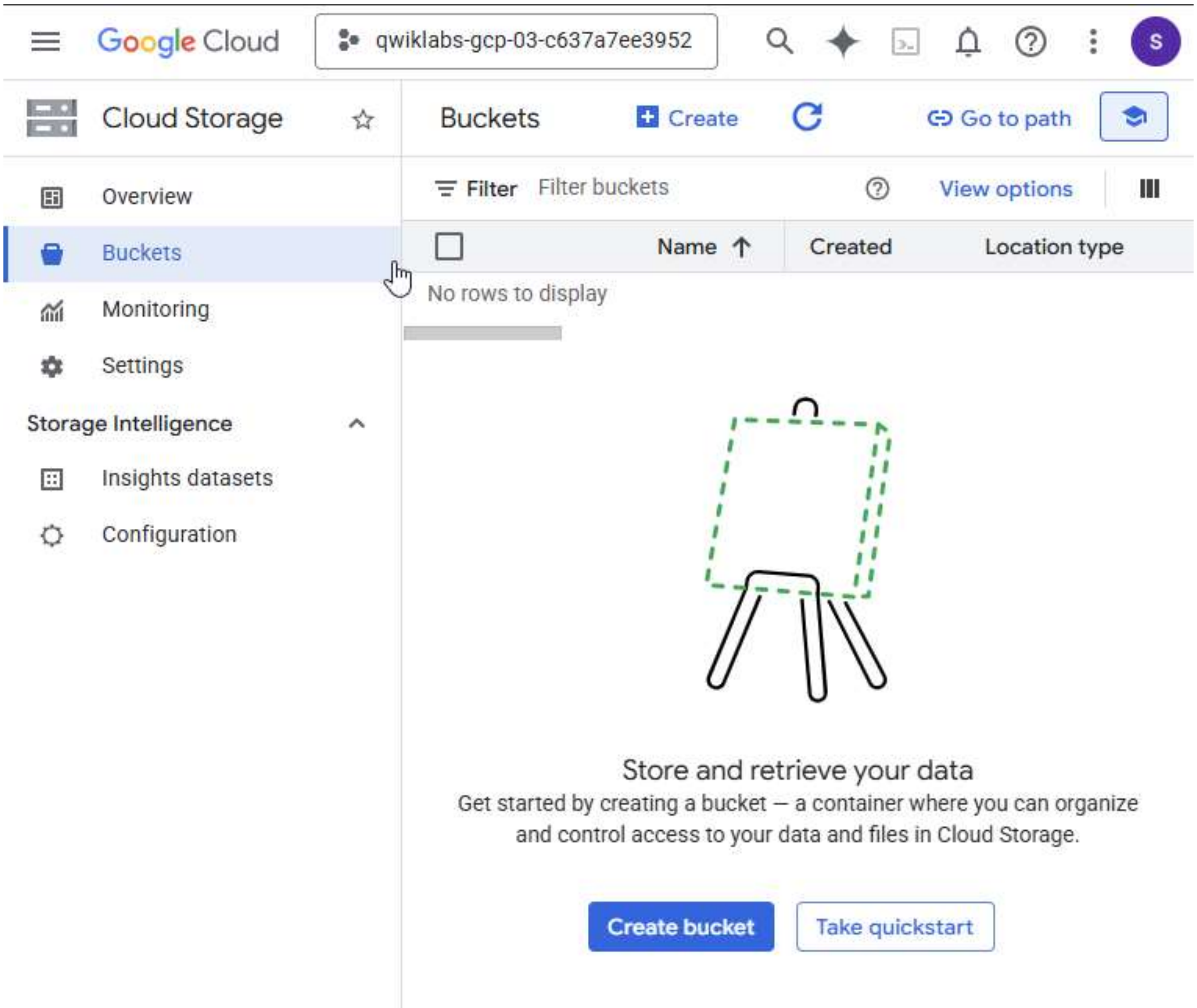
Task 2. Enable Private Google Access

VM instances that have no external IP addresses can use Private Google Access to reach external IP addresses of Google APIs and services. By default, Private Google Access is disabled on a VPC network.

Create a Cloud Storage bucket

Create a Cloud Storage bucket to test access to Google APIs and services.

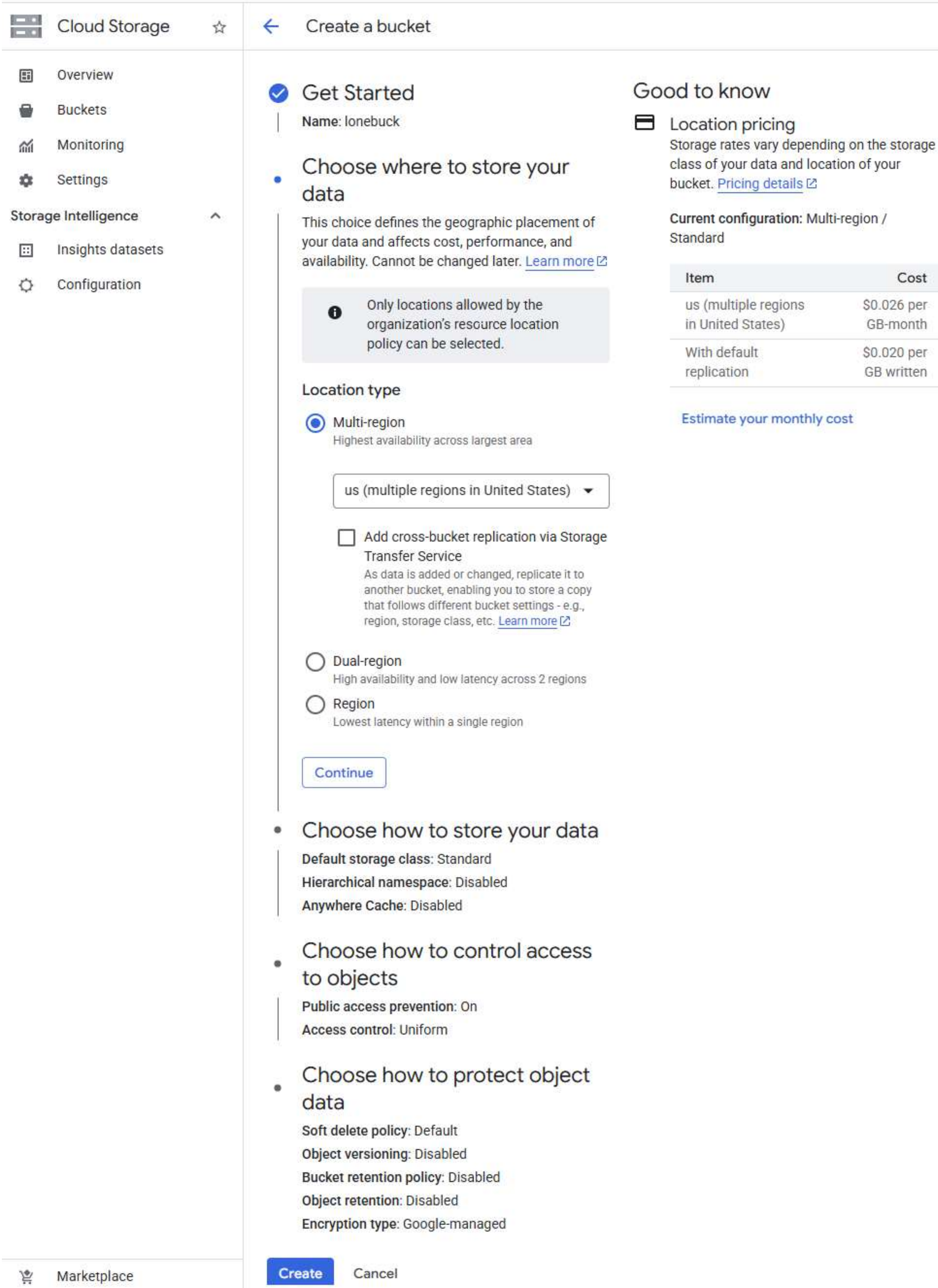
1. In the Google Cloud console, in the **Navigation menu** (☰), click **Cloud Storage > Buckets**.
2. Click **+Create**.



3. Specify the following, and leave the remaining settings as their defaults:

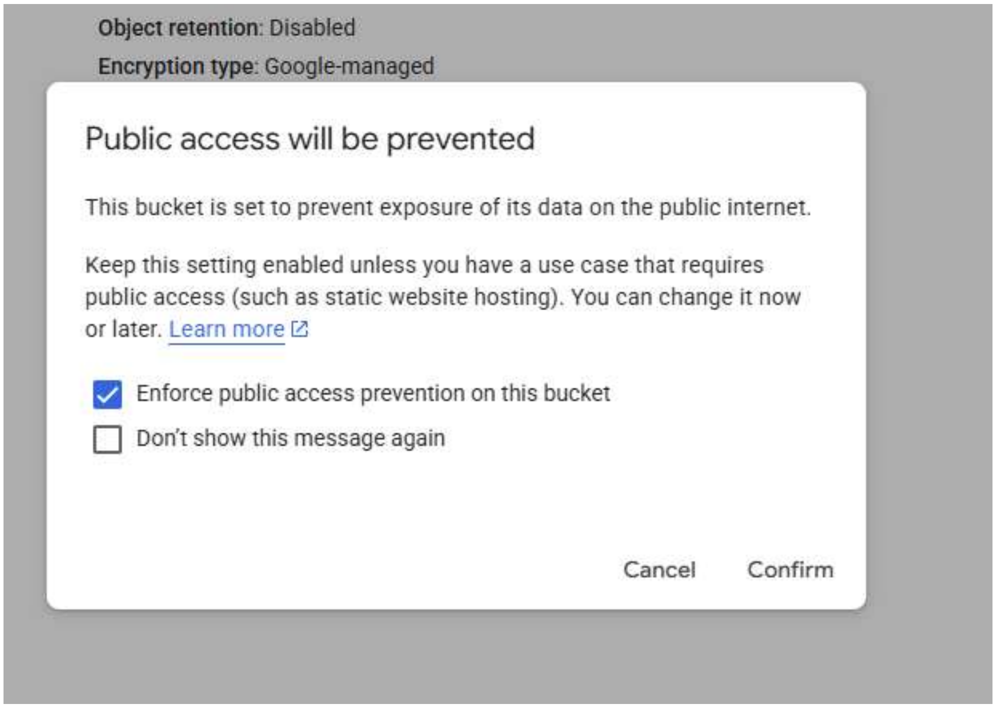
Property	**Value (type value or select option as specified)**
Name	*Enter a globally unique name*
Location type	Multi-region

4. Click **Create**.



Cloud Storage
Create a bucket
Overview
Buckets
Monitoring
Settings
Storage Intelligence
Insights datasets
Configuration
Marketplace
Get Started
Name: lonebuck
Choose where to store your data
This choice defines the geographic placement of your data and affects cost, performance, and availability. Cannot be changed later. Learn more
Only locations allowed by the organization's resource location policy can be selected.
Location type
Multi-region
Highest availability across largest area
us (multiple regions in United States)
Add cross-bucket replication via Storage Transfer Service
As data is added or changed, replicate it to another bucket, enabling you to store a copy that follows different bucket settings - e.g., region, storage class, etc. Learn more
Dual-region
High availability and low latency across 2 regions
Region
Lowest latency within a single region
Continue
Choose how to store your data
Default storage class: Standard
Hierarchical namespace: Disabled
Anywhere Cache: Disabled
Choose how to control access to objects
Public access prevention: On
Access control: Uniform
Choose how to protect object data
Soft delete policy: Default
Object versioning: Disabled
Bucket retention policy: Disabled
Object retention: Disabled
Encryption type: Google-managed
Create
Cancel
Good to know
Location pricing
Storage rates vary depending on the storage class of your data and location of your bucket. Pricing details
Current configuration: Multi-region / Standard
Item
Cost
us (multiple regions in United States)
$0.026 per GB-month
With default replication
$0.020 per GB written
Estimate your monthly cost

If prompted to enable public access prevention, ensure it is checked and cick **Confirm**. Note the name of your storage bucket.



5. Store the name of your bucket in an environment variable:
6. Verify it with echo:

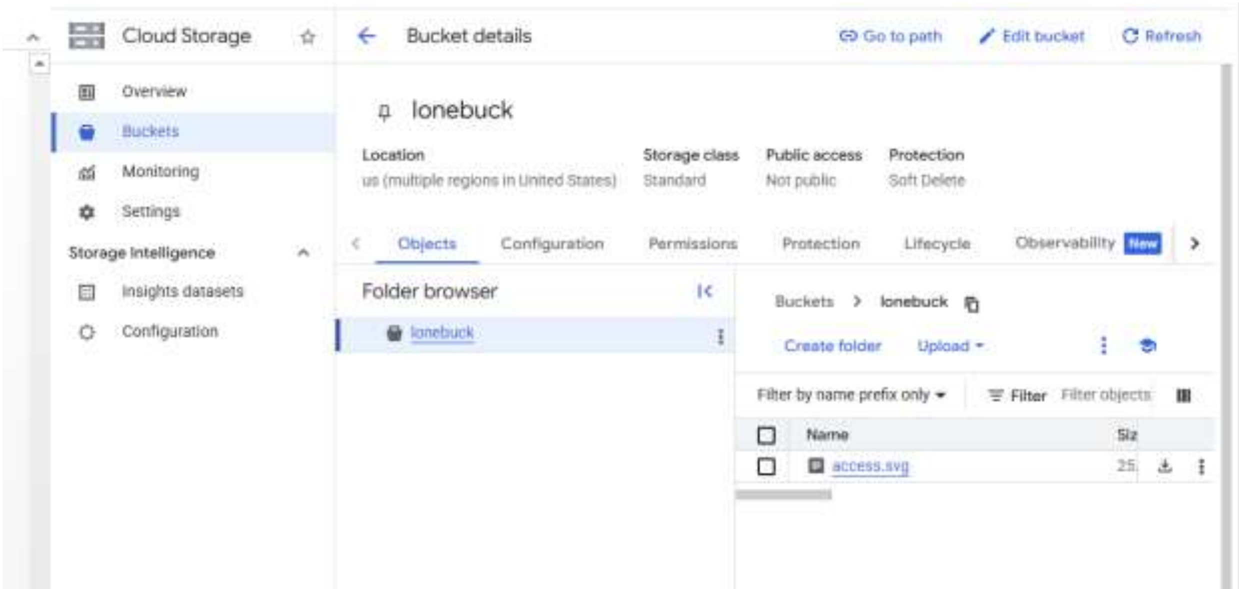
Copy an image file into your bucket

Copy an image from a public Cloud Storage bucket to your own bucket.

1. In Cloud Shell, run the following command:

2. In the Cloud console, click your bucket name to verify that the image was copied.

You can click on the name of the image in the Cloud console to view an example of how Private Google Access is implemented.



Access the image from your VM instance

Currently, which of your VM instances can access the image from your bucket?

☑

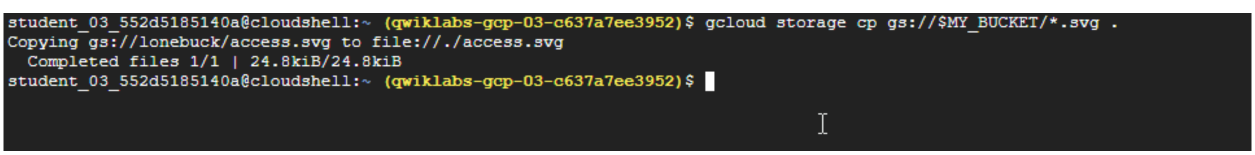
Cloud Shell

☐

vm-internal

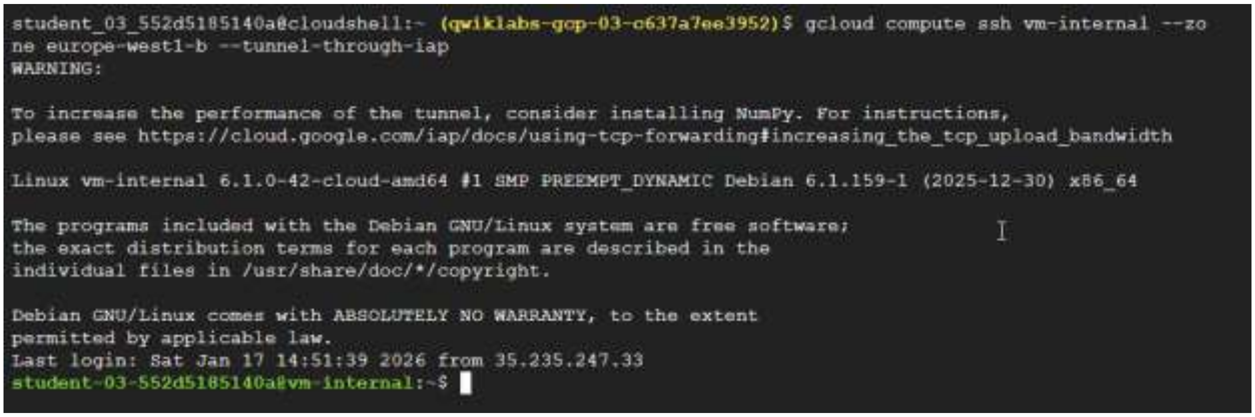
1. In Cloud Shell, to try to copy the image from your bucket, run the following command:

This should work because Cloud Shell has an external IP address!



2. To connect to vm-internal, run the following command:

3. If prompted, type **Y** to continue.



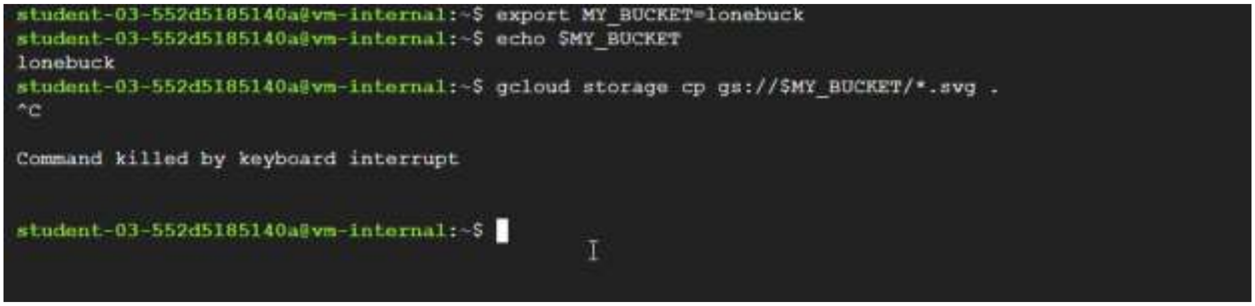
```
student_03_552d5185140a@cloudshell:~ (qwiklabs-gcp-03-c637a7ee3952)$ gcloud compute ssh vm-internal --zo
ne europe-west1-b --tunnel-through-iap
WARNING:

To increase the performance of the tunnel, consider installing NumPy. For instructions,
please see https://cloud.google.com/iap/docs/using-tcp-forwarding#increasing_the_tcp_upload_bandwidth

Linux vm-internal 6.1.0-42-cloud-amd64 #1 SMP PREEMPT_DYNAMIC Debian 6.1.159-1 (2025-12-30) x86_64

The programs included with the Debian GNU/Linux system are free software;
the exact distribution terms for each program are described in the
individual files in /usr/share/doc/*/copyright.

Debian GNU/Linux comes with ABSOLUTELY NO WARRANTY, to the extent
permitted by applicable law.
Last login: Sat Jan 17 14:51:39 2026 from 35.235.247.33
student-03-552d5185140a@vm-internal:~$
```

4. Store the name of your bucket in an environment variable:

5. Verify it with echo:
6. Try to copy the image to **vm-internal**, run the following command:

This should not work: vm-internal can only send traffic within the VPC network because Private Google Access is disabled (by default).

7. Press Ctrl+C to stop the request.



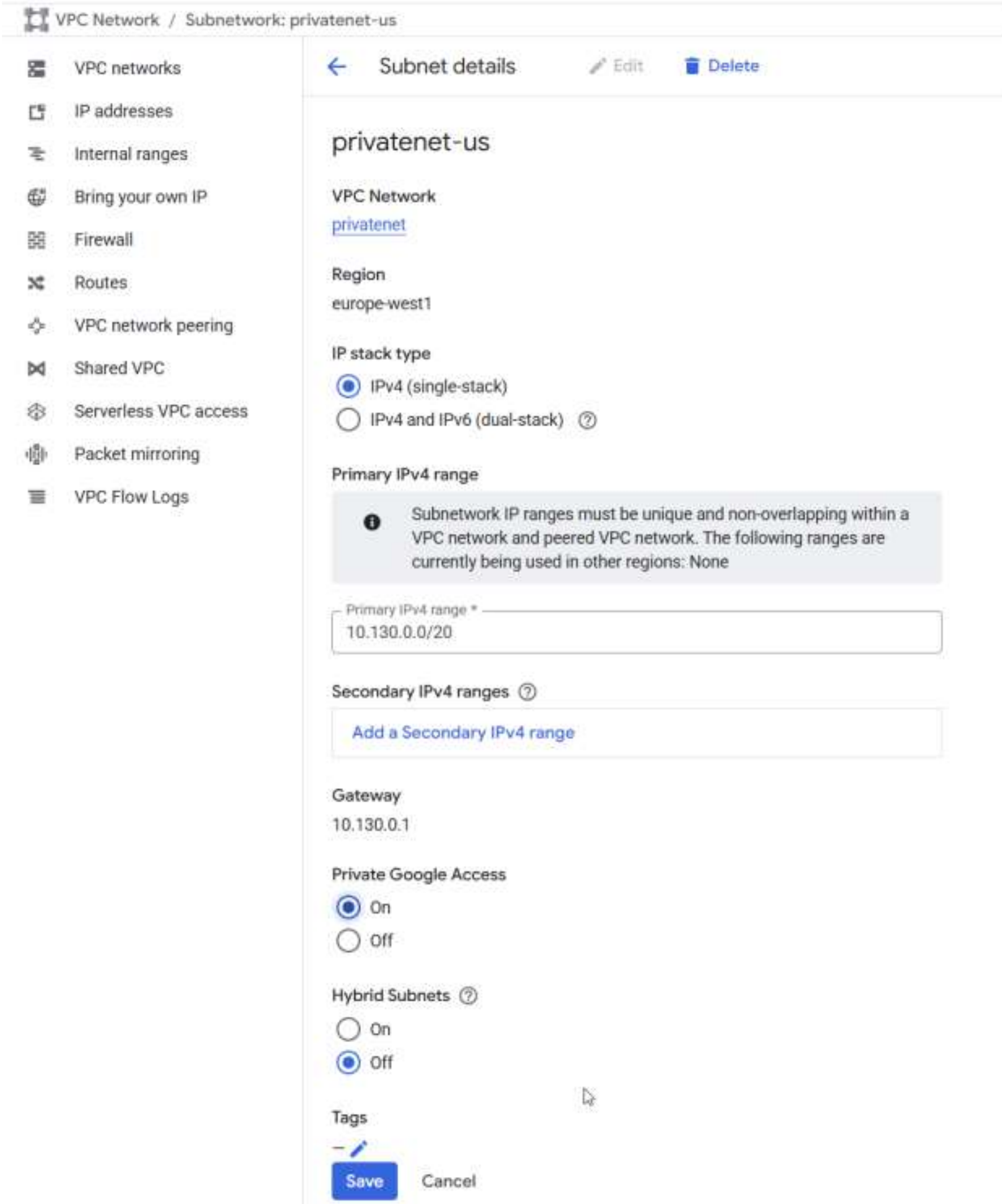
```
student-03-552d5185140a@vm-internal:~$ export MY_BUCKET=lonebuck
student-03-552d5185140a@vm-internal:~$ echo $MY_BUCKET
lonebuck
student-03-552d5185140a@vm-internal:~$ gcloud storage cp gs://$MY_BUCKET/*.svg .
^C

Command killed by keyboard interrupt

student-03-552d5185140a@vm-internal:~$
```

Enable Private Google Access

Private Google Access is enabled at the subnet level. When it is enabled, instances in the subnet that only have private IP addresses can send traffic to Google APIs and services through the default route (0.0.0.0/0) with a next hop to the default internet gateway.

1. In the Cloud console, in the **Navigation menu** (☰), click **VPC network > VPC networks**.
2. Click **privatenet** to open the network.
3. Click **Subnets**, and then click **privatenet-us**.
4. Click **Edit**.
5. For **Private Google access**, select **On**.
6. Click **Save**.

Note: *Enabling Private Google Access is as simple as selecting* ***On*** *within the subnet!*



7. Run the following command, in **Cloud Shell** for **vm-internal**, to try to copy the image to **vm-internal**.

This should work because vm-internal's subnet has Private Google Access enabled!

8. To return to your Cloud Shell instance, run the following command:

```
student-03-552d5185140a@vm-internal:~$ gcloud storage cp gs://$MY_BUCKET/*.svg .
Copying gs://lonebuck/access.svg to file://./access.svg
  Completed files 1/1 | 24.8kiB/24.8kiB
student-03-552d5185140a@vm-internal:~$ exit
logout
Connection to compute.8309808766384480571 closed.
student_03_552d5185140a@cloudshell:~ (qwiklabs-gcp-03-c637a7ee3952)$
```

Task 3. Configure a Cloud NAT gateway

Although vm-internal can now access certain Google APIs and services without an external IP address, the instance cannot access the internet for updates and patches. Configure a Cloud NAT gateway, which allows vm-internal to reach the internet.

Try to update the VM instances

1. In Cloud Shell, to try to re-synchronize the package index, run the following:

```
student_03_552d5185140a@cloudshell:~ (qwiklabs-gcp-03-c637a7ee3952)$ sudo apt-get update
********************************************************************************
You are running apt-get inside of Cloud Shell. Note that your Cloud Shell
machine is ephemeral and no system-wide change will persist beyond session end.

To suppress this warning, create an empty ~/.cloudshell/no-apt-get-warning file.
The command will automatically proceed in 5 seconds or on any key.

Visit https://cloud.google.com/shell/help for more information.
********************************************************************************
Hit:1 http://archive.ubuntu.com/ubuntu noble InRelease
Get:2 http://archive.ubuntu.com/ubuntu noble-updates InRelease [126 kB]
Get:3 http://archive.ubuntu.com/ubuntu noble-backports InRelease [126 kB]
Get:4 https://packages.cloud.google.com/apt gcsfuse-noble InRelease [1,227 B]
Get:5 https://apt.postgresql.org/pub/repos/apt noble-pgdg InRelease [107 kB]
Get:6 https://packages.cloud.google.com/apt cloud-sdk InRelease [1,620 B]
Get:7 https://ppa.launchpadcontent.net/dotnet/backports/ubuntu noble InRelease [24.1 kB]
Get:8 https://cli.github.com/packages stable InRelease [3,917 B]
Get:9 http://archive.ubuntu.com/ubuntu noble-updates/main amd64 Packages [2,142 kB]
Get:10 http://archive.ubuntu.com/ubuntu noble-updates/restricted amd64 Packages [3,077 kB]
Get:11 http://archive.ubuntu.com/ubuntu noble-updates/universe amd64 Packages [1,959 kB]
Get:12 https://packages.cloud.google.com/apt cloud-sdk/main amd64 Packages [4,414 kB]
Get:13 https://apt.postgresql.org/pub/repos/apt noble-pgdg/main amd64 Packages [579 kB]
Get:14 https://packages.cloud.google.com/apt cloud-sdk/main all Packages [1,911 kB]
Get:15 https://ppa.launchpadcontent.net/dotnet/backports/ubuntu noble/main amd64 Packages [6,476 B]
Get:16 https://cli.github.com/packages stable/main amd64 Packages [354 B]
Get:17 http://security.ubuntu.com/ubuntu noble-security InRelease [126 kB]
Get:18 https://download.docker.com/linux/ubuntu noble InRelease [48.5 kB]
Get:19 https://repo.mongodb.org/apt/ubuntu noble/mongodb-org/8.0 InRelease [3,005 B]
Get:20 http://security.ubuntu.com/ubuntu noble-security/universe amd64 Packages [1,191 kB]
Get:21 http://security.ubuntu.com/ubuntu noble-security/restricted amd64 Packages [2,919 kB]
Get:22 http://security.ubuntu.com/ubuntu noble-security/main amd64 Packages [1,769 kB]
Get:23 https://download.docker.com/linux/ubuntu noble/stable amd64 Packages [50.7 kB]
Get:24 https://repo.mongodb.org/apt/ubuntu noble/mongodb-org/8.0/multiverse amd64 Packages [58.3 kB]
Fetched 20.6 MB in 6s (3,518 kB/s)
Reading package lists... Done
student_03_552d5185140a@cloudshell:~ (qwiklabs-gcp-03-c637a7ee3952)$
```

2. To connect to vm-internal, run the following command:

3. If prompted, type Y to continue.

```
student_03_552d5185140a@cloudshell:~ (qwiklabs-gcp-03-c637a7ee3952)$ gcloud compute ssh vm-internal --zo
ne europe-west1-b --tunnel-through-iap
WARNING:

To increase the performance of the tunnel, consider installing NumPy. For instructions,
please see https://cloud.google.com/iap/docs/using-tcp-forwarding#increasing_the_tcp_upload_bandwidth

Linux vm-internal 6.1.0-42-cloud-amd64 #1 SMP PREEMPT_DYNAMIC Debian 6.1.159-1 (2025-12-30) x86_64

The programs included with the Debian GNU/Linux system are free software;
the exact distribution terms for each program are described in the
individual files in /usr/share/doc/*/copyright.

Debian GNU/Linux comes with ABSOLUTELY NO WARRANTY, to the extent
permitted by applicable law.
Last login: Sat Jan 17 15:07:33 2026 from 35.235.247.33
student-03-552d5185140a@vm-internal:~$
```

4. To try to re-synchronize the package index of vm-internal, run the following command:

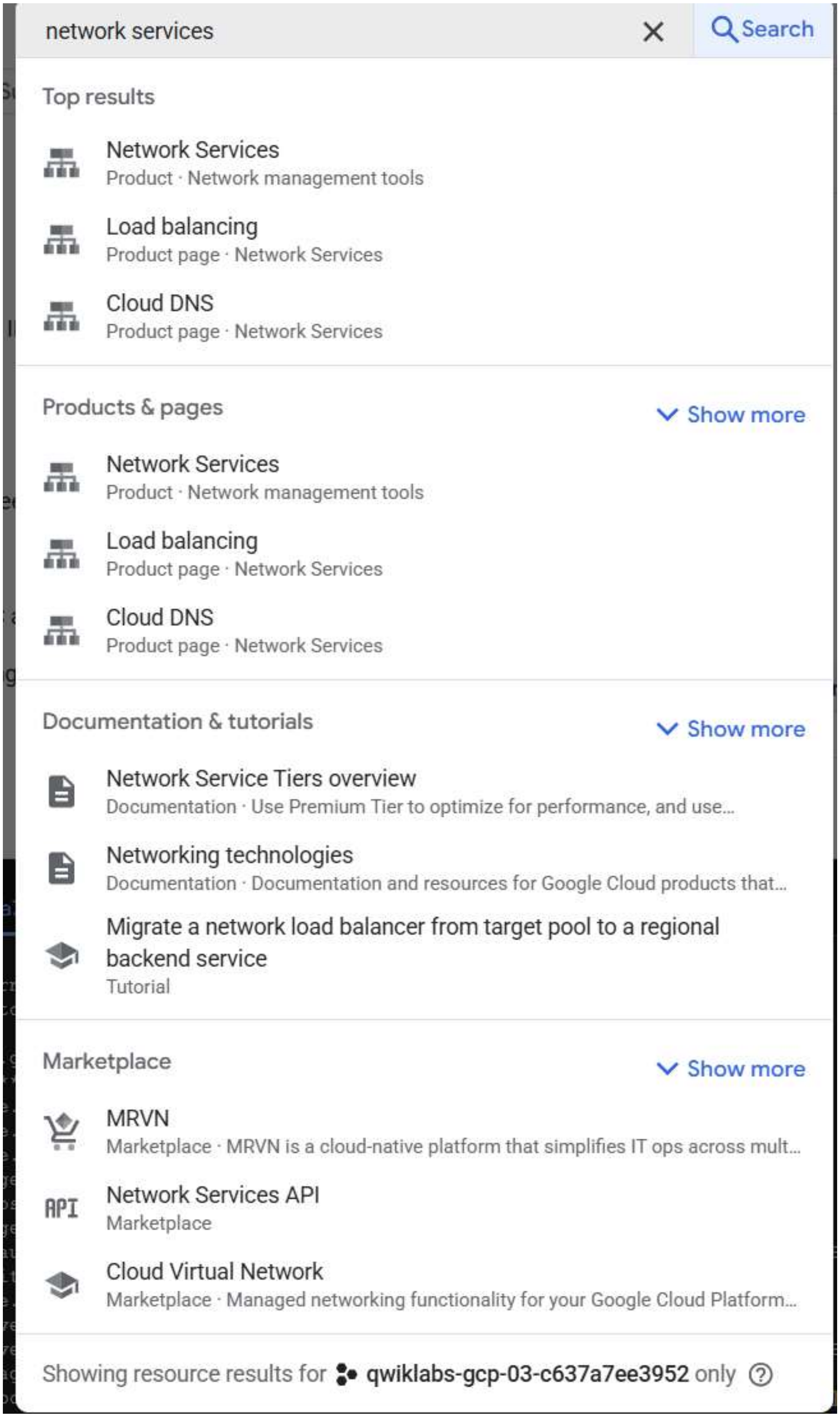
This should only work for Google Cloud packages because vm-internal only has access to Google APIs and services!

```
student-03-552d5185140a@vm-internal:~$ sudo apt-get update
Reading package lists... Done
E: Could not get lock /var/lib/apt/lists/lock. It is held by process 1357 (apt-get)
N: Be aware that removing the lock file is not a solution and may break your system.
E: Unable to lock directory /var/lib/apt/lists/
student-03-552d5185140a@vm-internal:~$
```

Configure a Cloud NAT gateway

Cloud NAT is a regional resource. You can configure it to allow traffic from all ranges of all subnets in a region, from specific subnets in the region only, or from specific primary and secondary CIDR ranges only.

1. On the Google Cloud console title bar, type **Network services** in the **Search** field, then click **Network services** in the **Products & Page** section.


network services
Search
Top results
Network Services
Product · Network management tools
Load balancing
Product page · Network Services
Cloud DNS
Product page · Network Services
Products & pages
Show more
Network Services
Product · Network management tools
Load balancing
Product page · Network Services
Cloud DNS
Product page · Network Services
Documentation & tutorials
Show more
Network Service Tiers overview
Documentation · Use Premium Tier to optimize for performance, and use...
Networking technologies
Documentation · Documentation and resources for Google Cloud products that...
Migrate a network load balancer from target pool to a regional backend service
Tutorial
Marketplace
Show more
MRVN
Marketplace · MRVN is a cloud-native platform that simplifies IT ops across mult...
Network Services API
Marketplace
Cloud Virtual Network
Marketplace · Managed networking functionality for your Google Cloud Platform...
Showing resource results for qwiklabs-gcp-03-c637a7ee3952 only

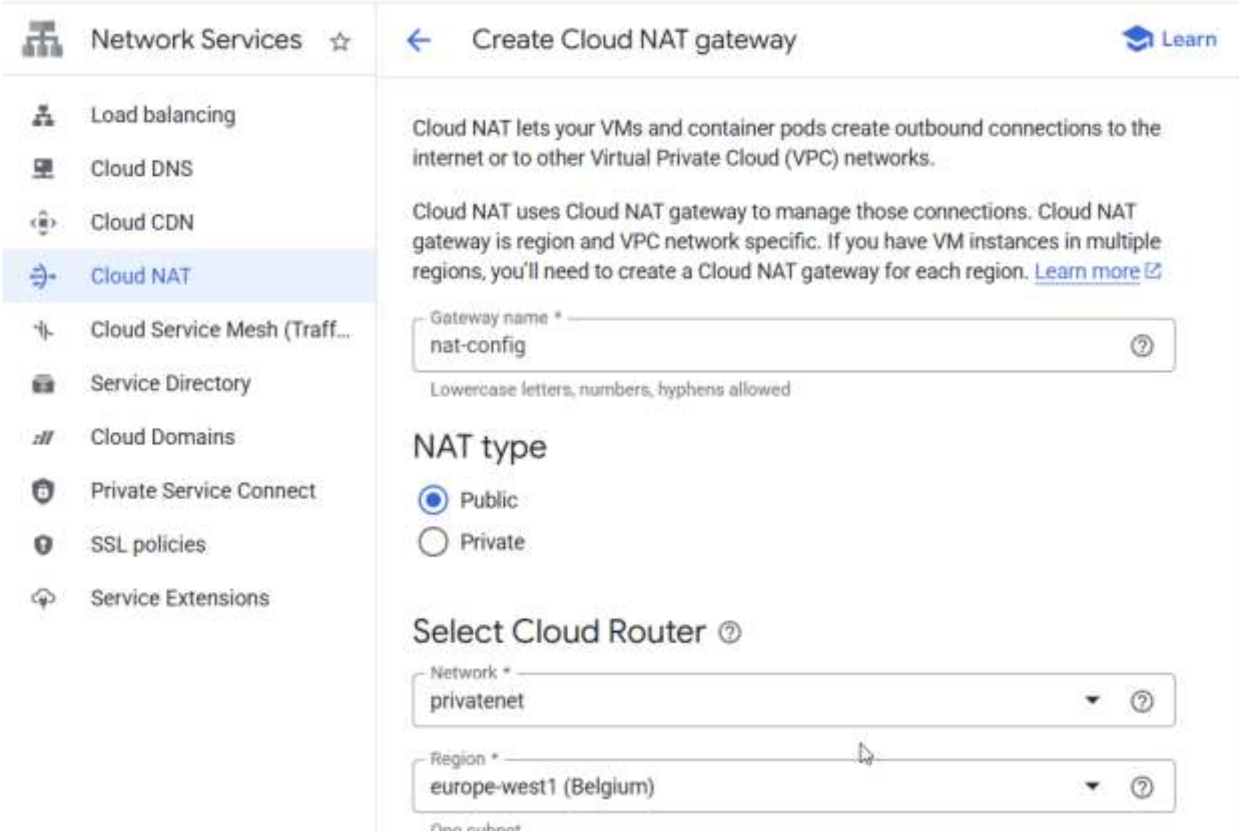
2. On the **Network service** page, click **Pin** next to Network services.

3. Click **Cloud NAT**.

4. Click **Get started** to configure a NAT gateway.

5. Specify the following:

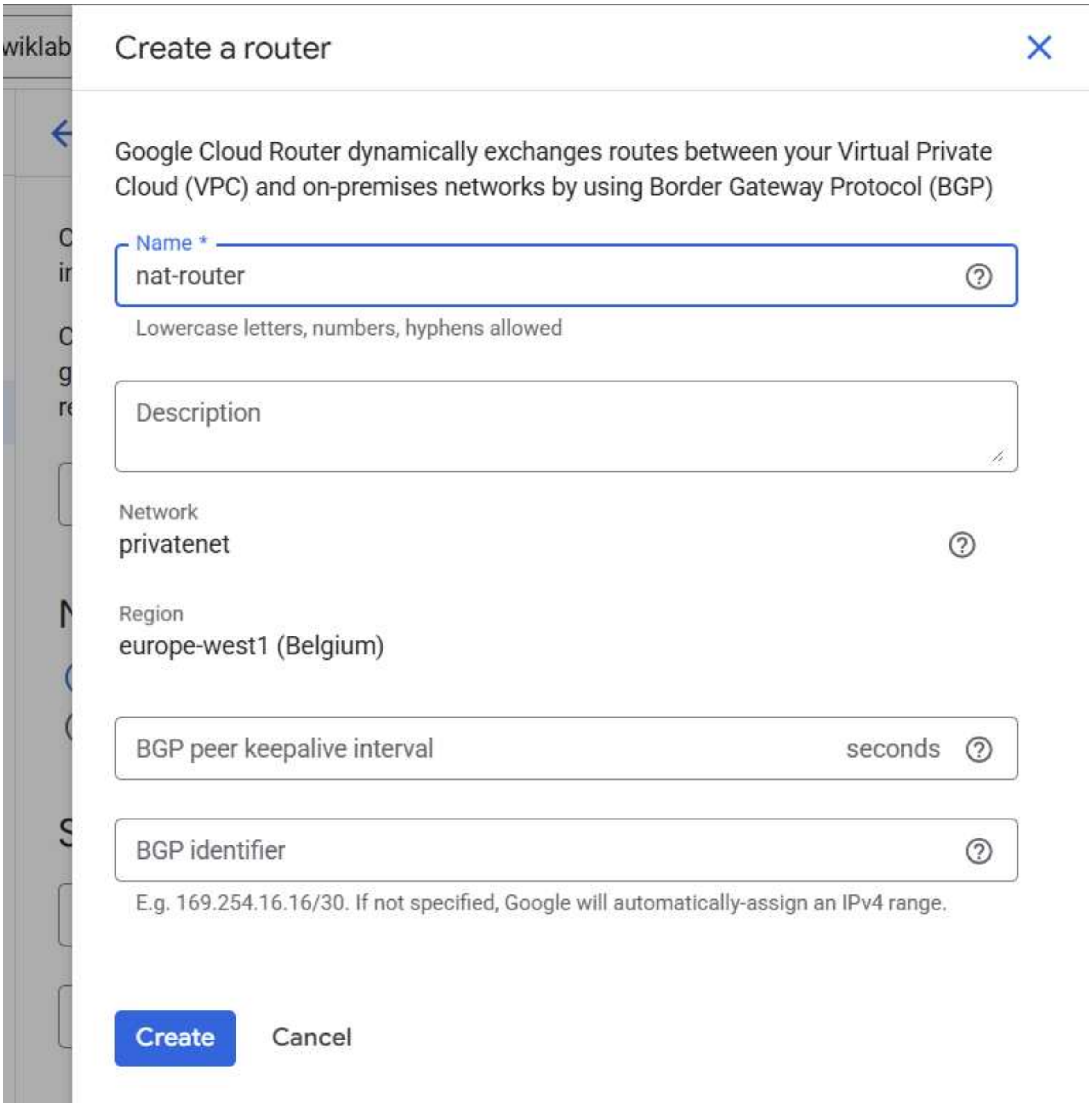
Property	**Value (type value or select option as specified)**
Gateway name	nat-config
Network	privatenet
Region	



6. For **Cloud Router**, select **Create new router**.

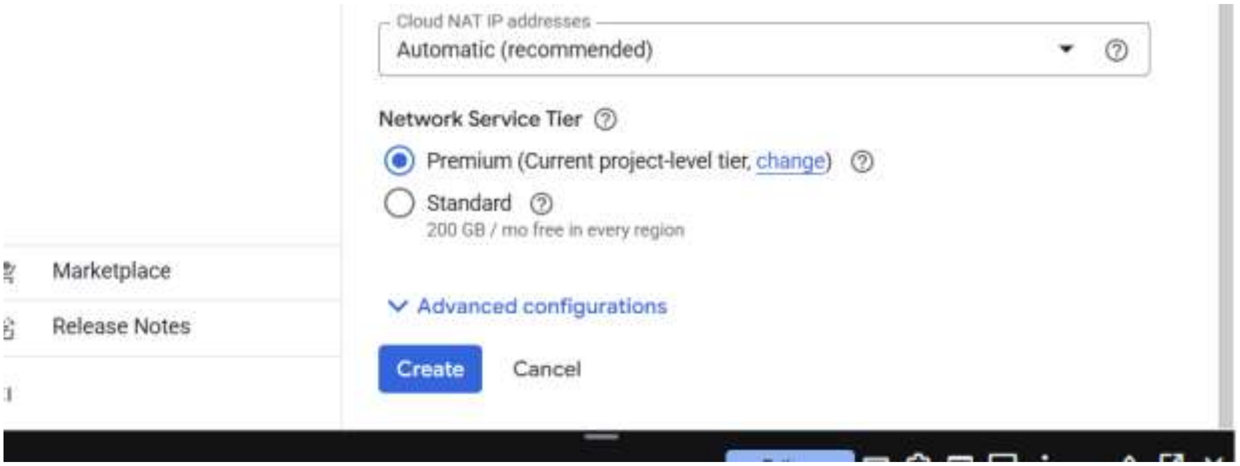
7. For **Name**, type **nat-router**

8. Click **Create**.

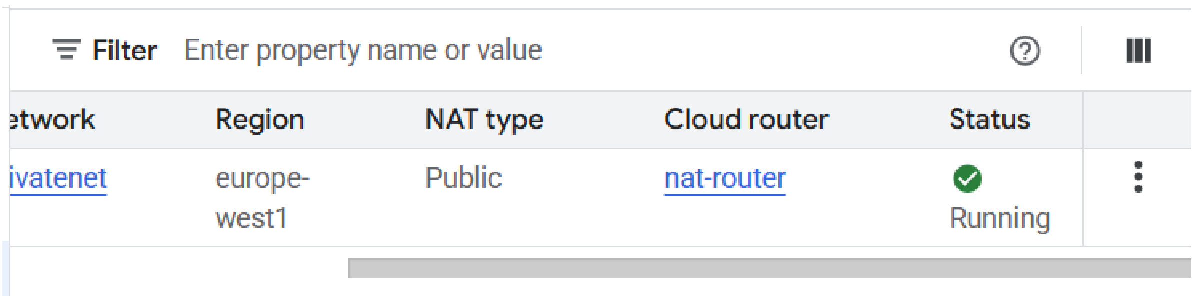


*Note: The **NAT mapping** section allows you to choose the subnets to map to the NAT gateway. You can also manually assign static IP addresses that should be used when performing NAT. Do not change the NAT mapping configuration in this lab.*

9. Click **Create**.



10. Wait for the gateway's status to change to **Running**.



Filter Enter property name or value

etwork	Region	NAT type	Cloud router	Status
ivatenet	europe-west1	Public	nat-router	Running

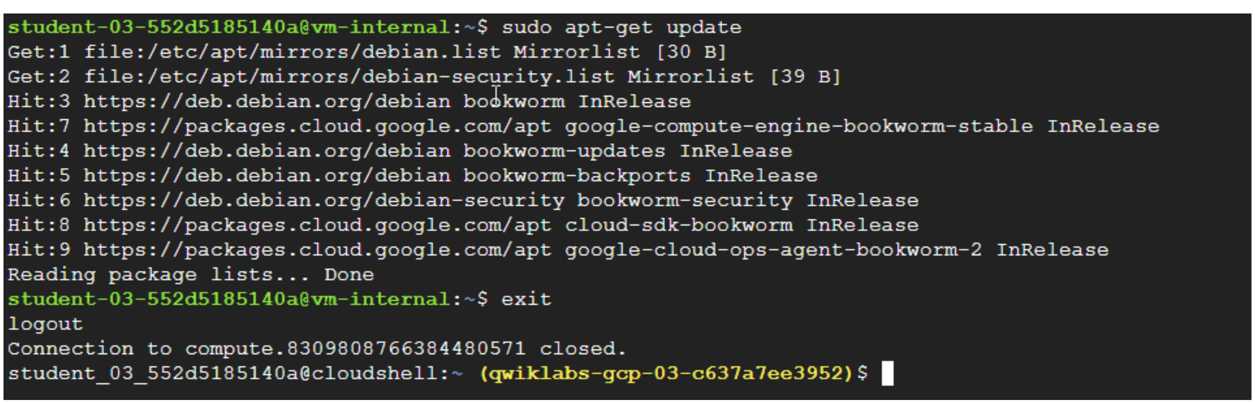
Verify the Cloud NAT gateway

It may take up to 3 minutes for the NAT configuration to propagate to the VM, so wait at least a minute before trying to access the internet again.

1. In Cloud Shell for vm-internal, to try to re-synchronize the package index of vm-internal, run the following command:

This should work because ***vm-internal*** *is using the NAT gateway!*

2. To return to your Cloud Shell instance, run the following command:



```
student-03-552d5185140a@vm-internal:~$ sudo apt-get update
Get:1 file:/etc/apt/mirrors/debian.list Mirrorlist [30 B]
Get:2 file:/etc/apt/mirrors/debian-security.list Mirrorlist [39 B]
Hit:3 https://deb.debian.org/debian bookworm InRelease
Hit:7 https://packages.cloud.google.com/apt google-compute-engine-bookworm-stable InRelease
Hit:4 https://deb.debian.org/debian bookworm-updates InRelease
Hit:5 https://deb.debian.org/debian bookworm-backports InRelease
Hit:6 https://deb.debian.org/debian-security bookworm-security InRelease
Hit:8 https://packages.cloud.google.com/apt cloud-sdk-bookworm InRelease
Hit:9 https://packages.cloud.google.com/apt google-cloud-ops-agent-bookworm-2 InRelease
Reading package lists... Done
student-03-552d5185140a@vm-internal:~$ exit
logout
Connection to compute.8309808766384480571 closed.
student_03_552d5185140a@cloudshell:~ (qwiklabs-gcp-03-c637a7ee3952)$
```

Note: The Cloud NAT gateway implements outbound NAT, but not inbound NAT. In other words, hosts outside of your VPC network can only respond to connections initiated by your instances; they cannot initiate their own, new connections to your instances via NAT.

Task 4. Configure and view logs with Cloud NAT Logging

Cloud NAT logging allows you to log NAT connections and errors. When Cloud NAT logging is enabled, one log entry can be generated for each of the following scenarios:

- When a network connection using NAT is created.
- When a packet is dropped because no port was available for NAT.

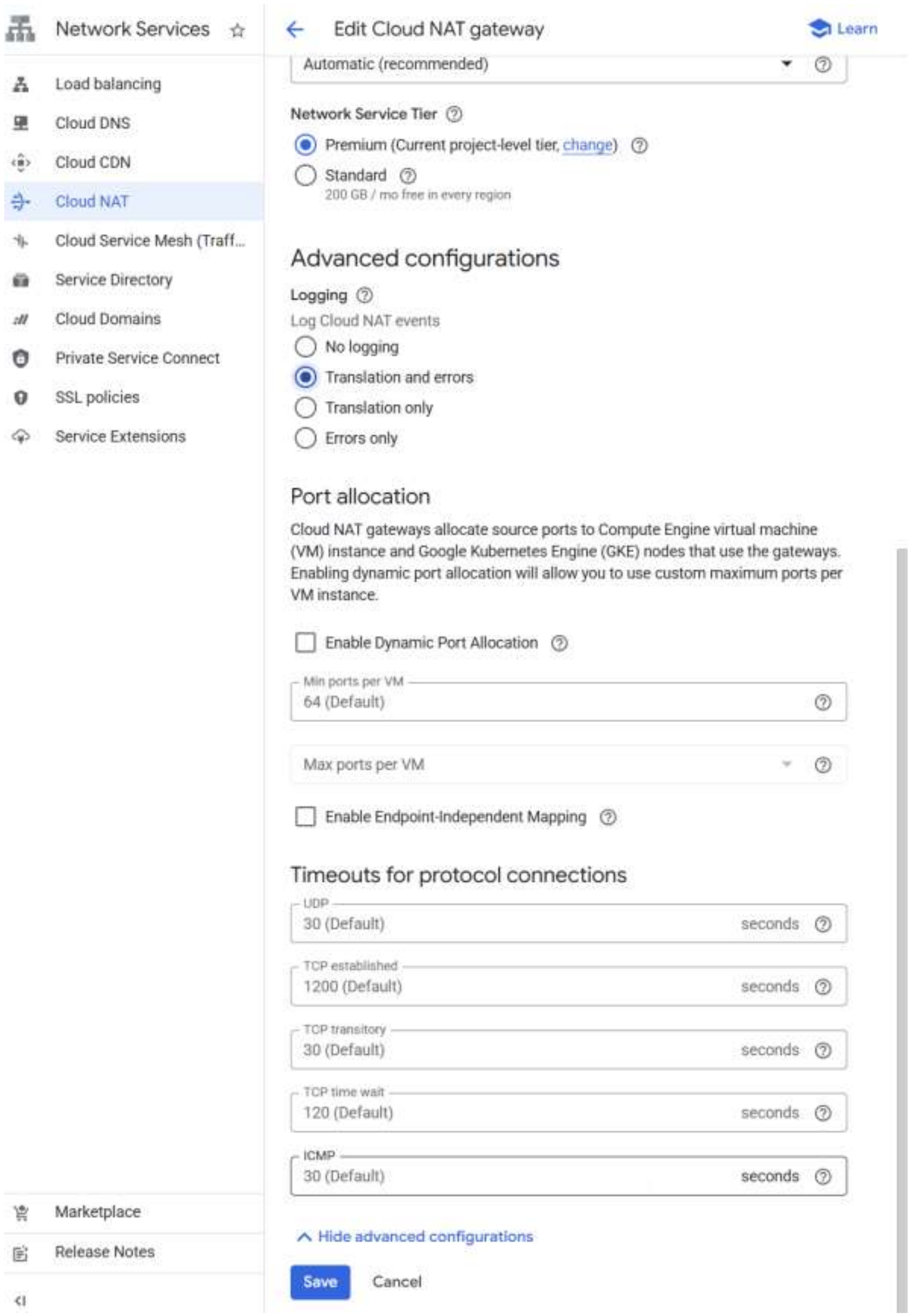
You can opt to log both kinds of events, or just one or the other. Created logs are sent to Cloud Logging.

Enabling logging

If logging is enabled, all collected logs are sent to Cloud Logging by default. You can filter these so that only certain logs are sent.

You can also specify these values when you create a NAT gateway or by editing one after it has been created. The following directions show how to enable logging for an existing NAT gateway.

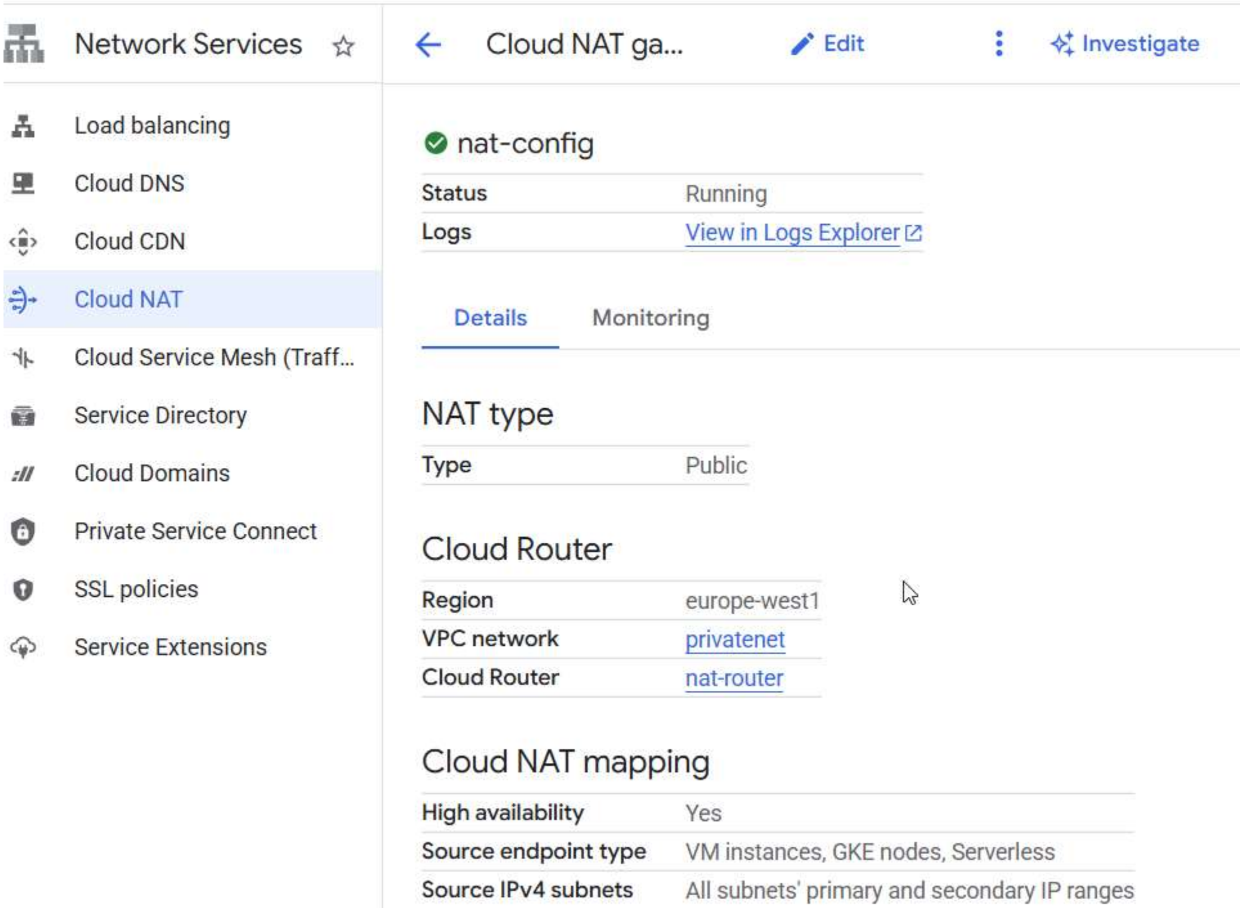
1. In the Google Cloud console, in the **Navigation menu** (☰), click **Network services** > **Cloud NAT**.
2. Click on the nat-config gateway and then click **Edit**.
3. Click the **Advanced configurations** dropdown to open that section.
4. For **Logging**, select **Translation and errors** and then click **Save**.


Network Services
Load balancing
Cloud DNS
Cloud CDN
Cloud NAT
Cloud Service Mesh (Traff...
Service Directory
Cloud Domains
Private Service Connect
SSL policies
Service Extensions
Marketplace
Release Notes
Edit Cloud NAT gateway
Learn
Automatic (recommended)
Network Service Tier
Premium (Current project-level tier, change)
Standard
200 GB / mo free in every region
Advanced configurations
Logging
Log Cloud NAT events
No logging
Translation and errors
Translation only
Errors only
Port allocation
Cloud NAT gateways allocate source ports to Compute Engine virtual machine (VM) instance and Google Kubernetes Engine (GKE) nodes that use the gateways. Enabling dynamic port allocation will allow you to use custom maximum ports per VM instance.
Enable Dynamic Port Allocation
Min ports per VM
64 (Default)
Max ports per VM
Enable Endpoint-Independent Mapping
Timeouts for protocol connections
UDP
30 (Default)
seconds
TCP established
1200 (Default)
seconds
TCP transitory
30 (Default)
seconds
TCP time wait
120 (Default)
seconds
ICMP
30 (Default)
seconds
Hide advanced configurations
Save
Cancel

NAT logging in Cloud Logging

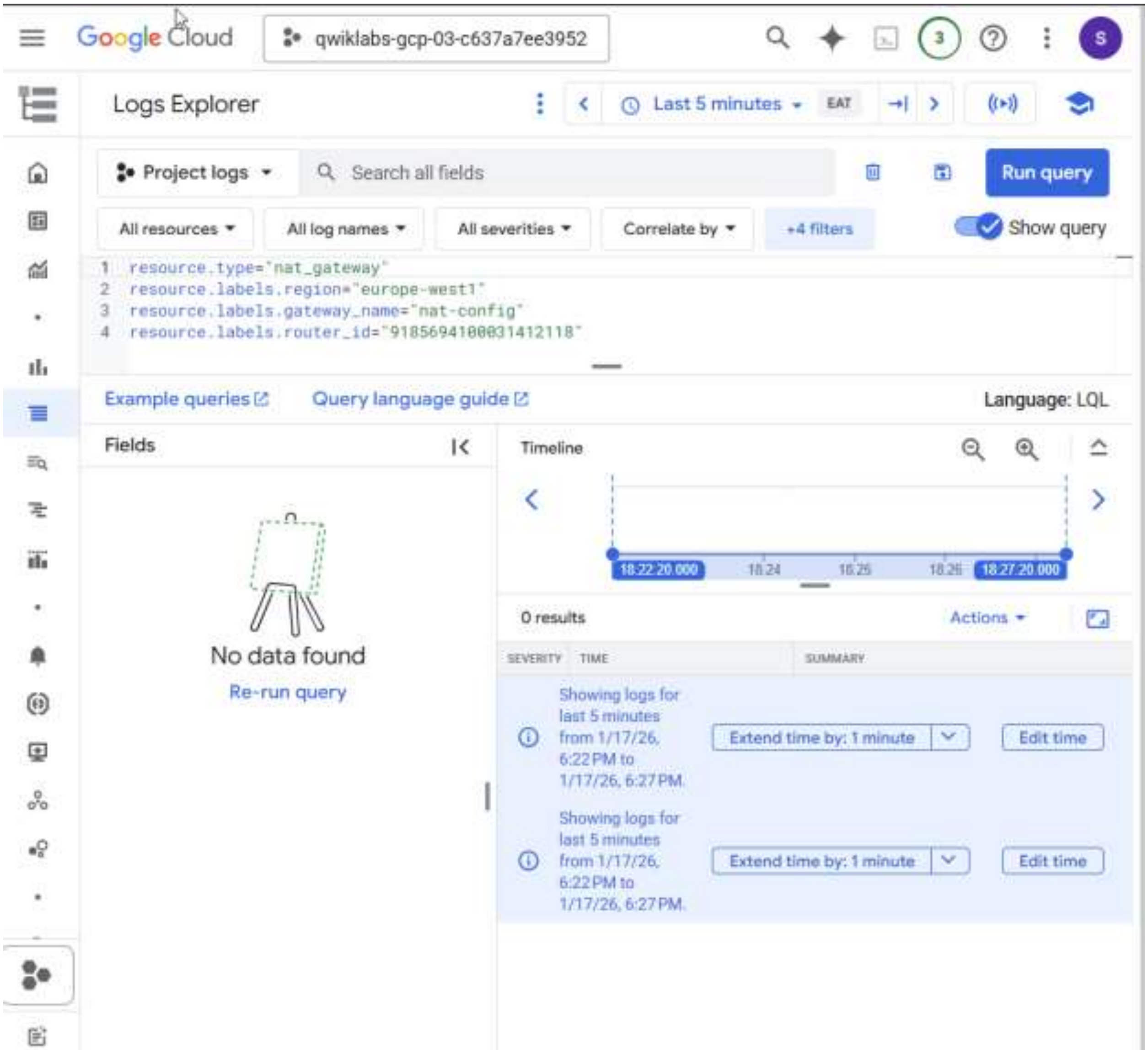
Now that you have set up Cloud NAT logging for the nat-config gateway, let's find out where we can view our logs.

1. Click on nat-config to expose its details. Then click on the **View in Logs Explorer**.



2. This will open a new tab with **Logs Explorer**.

You will see that there aren't any logs yet—that's because we just enabled this feature for the gateway.



***Note:** Keep this tab open and return to your other Google Cloud console tab.*

Generating logs

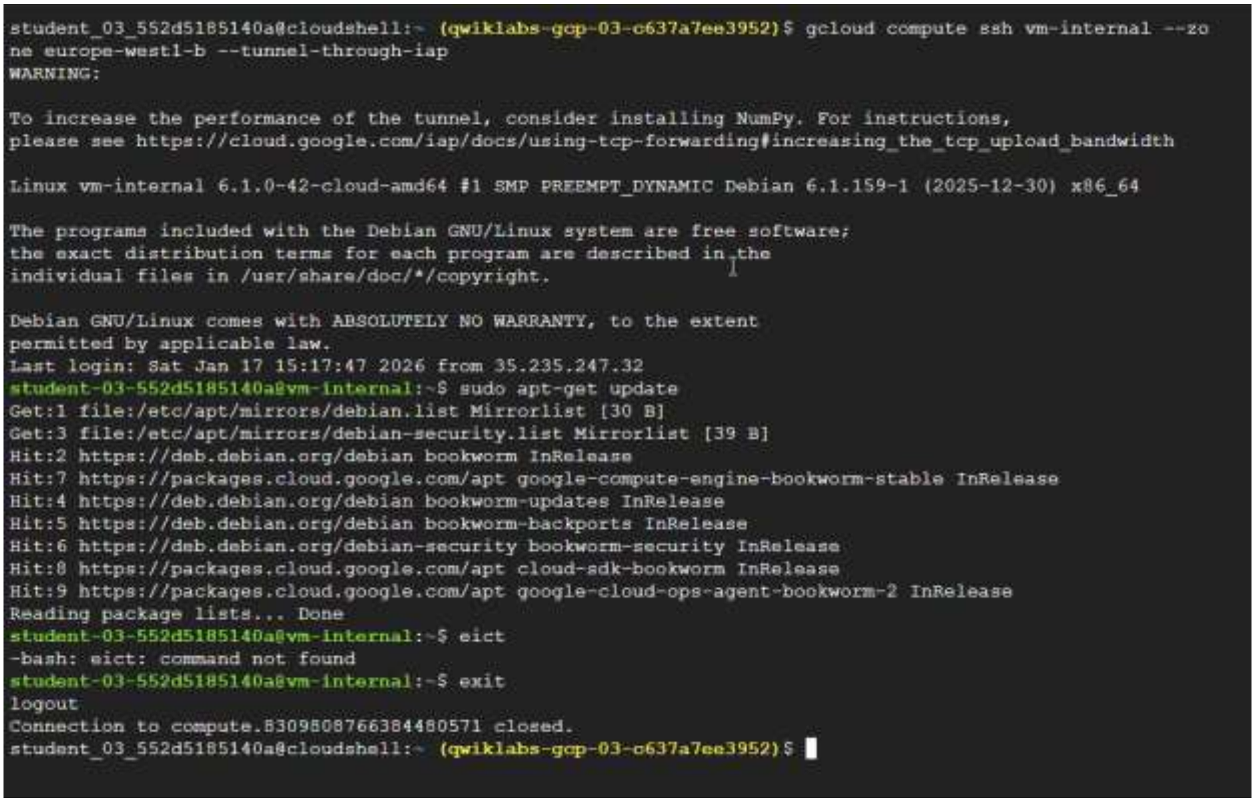
As a reminder, Cloud NAT logs are generated for the following sequences:

- When a network connection using NAT is created.
- When a packet is dropped because no port was available for NAT.

Let's connect the host to the internal VM again to see if any logs are generated.

1. In **Cloud Shell** for **vm-internal**, to try to re-synchronize the package index of **vm-internal**, run the following command:

2. If prompted, type **Y** to continue.
3. Try to re-synchronize the package index of **vm-internal** by running the following:
4. To return to your Cloud Shell instance, run the following command:



```
student_03_552d5185140a@cloudshell:~ (qwiklabs-gcp-03-c637a7ee3952)$ gcloud compute ssh vm-internal --zo
ne europe-west1-b --tunnel-through-iap
WARNING:

To increase the performance of the tunnel, consider installing NumPy. For instructions,
please see https://cloud.google.com/iap/docs/using-tcp-forwarding#increasing_the_tcp_upload_bandwidth

Linux vm-internal 6.1.0-42-cloud-amd64 #1 SMP PREEMPT_DYNAMIC Debian 6.1.159-1 (2025-12-30) x86_64

The programs included with the Debian GNU/Linux system are free software;
the exact distribution terms for each program are described in the
individual files in /usr/share/doc/*/copyright.

Debian GNU/Linux comes with ABSOLUTELY NO WARRANTY, to the extent
permitted by applicable law.
Last login: Sat Jan 17 15:17:47 2026 from 35.235.247.32
student-03-552d5185140a@vm-internal:~$ sudo apt-get update
Get:1 file:/etc/apt/mirrors/debian.list Mirrorlist [30 B]
Get:3 file:/etc/apt/mirrors/debian-security.list Mirrorlist [39 B]
Hit:2 https://deb.debian.org/debian bookworm InRelease
Hit:7 https://packages.cloud.google.com/apt google-compute-engine-bookworm-stable InRelease
Hit:4 https://deb.debian.org/debian bookworm-updates InRelease
Hit:5 https://deb.debian.org/debian bookworm-backports InRelease
Hit:6 https://deb.debian.org/debian-security bookworm-security InRelease
Hit:8 https://packages.cloud.google.com/apt cloud-sdk-bookworm InRelease
Hit:9 https://packages.cloud.google.com/apt google-cloud-ops-agent-bookworm-2 InRelease
Reading package lists... Done
student-03-552d5185140a@vm-internal:~$ eict
-bash: eict: command not found
student-03-552d5185140a@vm-internal:~$ exit
logout
Connection to compute.8309808766384480571 closed.
student_03_552d5185140a@cloudshell:~ (qwiklabs-gcp-03-c637a7ee3952)$
```

Let's see if opening up this connection revealed anything new in our logs.

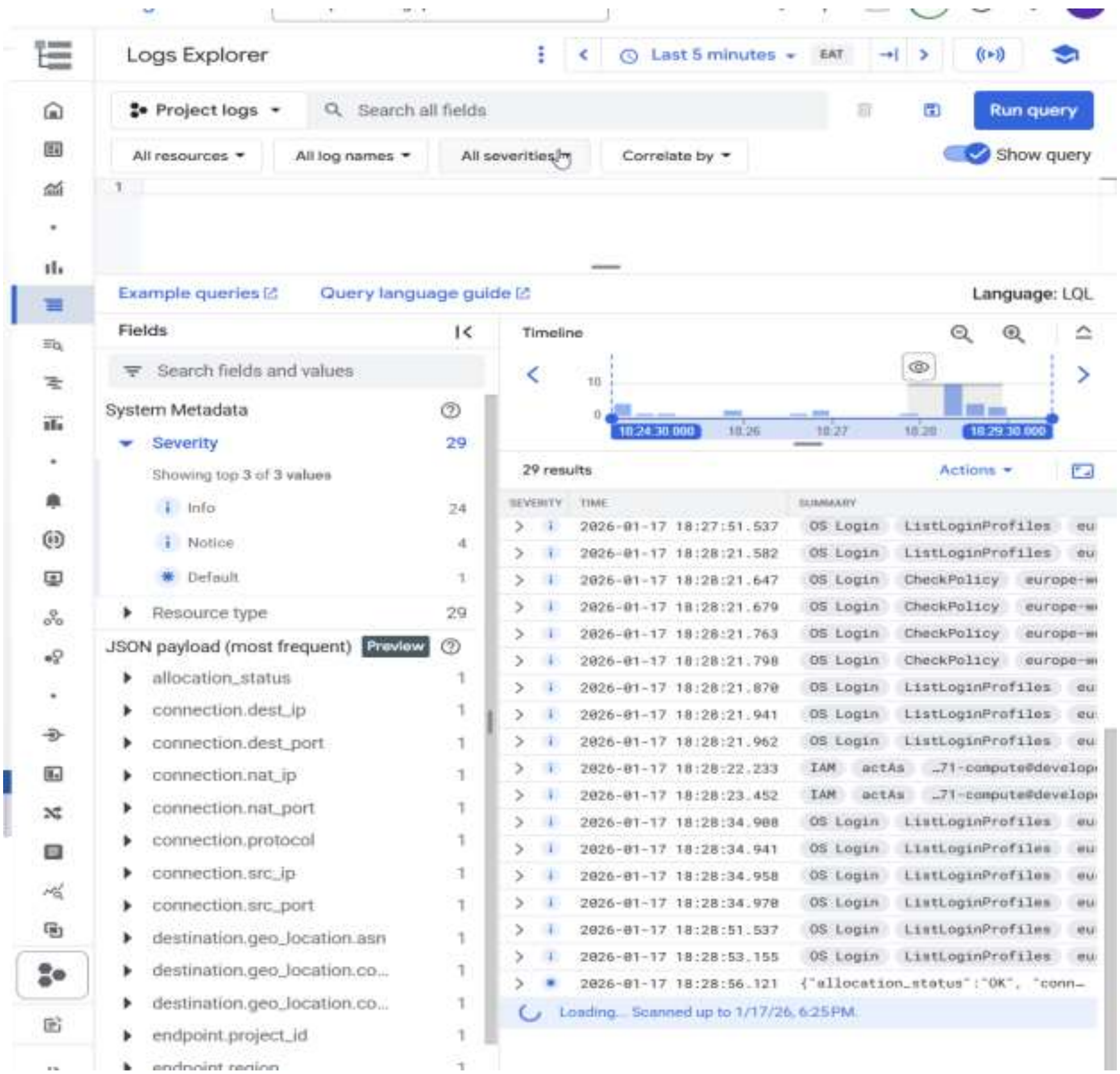
Viewing Logs

- Return to your Logs Explorer tab, and in the navigation menu, click **Logs Explorer**.

You should see two new logs that were generated after connecting to the internal VM.

Note: *You may need to wait for a few minutes. If you are still unable to see the logs, repeat step 1 to step 4, from the* ***Generating logs*** *section, and then refresh the logging page.*

As we see, the logs give us details on the VPC network we connected to and the connection method we used. Feel free to expand different labels and details.



Task 5. Review

I created **vm-internal**, an instance with no external IP address, and connected to it securely using an IAP tunnel. Then I enabled Private Google Access, configured a NAT gateway, and verified that **vm-internal** can access Google APIs and services and other public IP addresses.

VM instances without external IP addresses are isolated from external networks. Using Cloud NAT, these instances can access the internet for updates and patches, and in some cases, for bootstrapping. As a managed service, Cloud NAT provides high availability without user management and intervention.

IAP uses your existing project roles and permissions when you connect to VM instances. By default, instance owners are the only users that have the **IAP Secured Tunnel User** role.